Medical Laboratory Techniques

Level III

TVT Curriculum Based on Version 4

Occupational Standard

January, 2024

Addis Ababa, Ethiopia

Preface

The reformed TVET-System is an outcome-based system. It utilizes the needs of the labor market and occupational requirements from the world of work as the benchmark and standard for TVET delivery. The requirements from the world of work are analyzed and documented taking into account international benchmarking as occupational standards (OS).

In the reformed TVET-System, curricula and curriculum development play an important role with regard to quality driven TVET-Delivery. Curricula help to facilitate the learning process in a way, that trainees acquire the set of occupational competences (skills, knowledge and attitude) required at the working place and defined in the occupational standards (OS). Responsibility for Curriculum Development will be given to the Regional TVET-Authorities and TVET-Providers.

This curriculum has been developed by a group of experts from different Regional TVET-Authorities based on the occupational standard for Medical Laboratory Techniques Level III . It has the character of a model curriculum and is an example on how to transform the occupational requirements as defined in the respective occupational standard into an adequate curriculum.

The curriculum development process has been actively supported and facilitated by the Ministry of Health and Ministry of labor and skill

TVET-Program Design

1.1. TVET-Program Title: Medical Laboratory Techniques Level III

1.2. TVET-Program Description

The Program is designed to develop the necessary knowledge, skills and attitude of the learners to the standard required by the occupation. The contents of this program are in line with the occupational standard. Learners who successfully completed the Program will be qualified to work as a Medical Laboratory Technicians with competencies elaborated in the respective OS. Graduates of the program will have the required qualification to work in the health sector in the field of Medical Laboratory.

The prime objective of this training program is to equip the learners with the identified competences specified in the OS. Graduates are therefore expected to Provide First Aid and Emergency Response, Apply Infection Prevention Techniques and Workplace OHS, Provide Motivated Competent and Compassionate service, perform equipment handling and maintenance, Collect and Process Medical Samples, Prepare Laboratory Solutions, Apply Computer and Mobile Health Technology, Perform Urine and Body Fluid analysis, Perform Parasitological Examination, Apply 5S Procedures, Perform Community Mobilization and Provide Health Education and Apply basic health statistics and health survey in accordance with the performance criteria described in the OS.

1.3. TVET-Program Learning Outcomes

The expected outputs of this program are the acquisition and implementation of the following units of competences:

HLT MLT3 01 1224 Provide Motivated Competent and Compassionate service

HLT MLT3 02 1224 Apply Infection Prevention Techniques and Workplace OHS

HLT MLT3 03 1224 Provide First Aid and Emergency Response

HLT MLT3 05 1224 Perform equipment handling and maintenance

HLT MLT3 06 1224 Prepare Laboratory Solutions

HLT MLT3 04 1224 Collect and Process Medical Samples

HLT MLT3 09 1224 Apply Computer and Mobile Health Technology

HLT MLT3 07 1224 Perform Parasitological Examination

HLT MLT3 08 1224 Perform Urine and Body Fluid analysis

HLT MLT3 10 1224 Apply basic health statistics and health survey

HLT MLT3 11 1224 Perform Community Mobilization and Provide Health Education

HLT MLT3 12 1224 Apply 5S Procedures

Duration of the TVET-Program

The Program will have duration of 1738 hours including the on-the-job practice or cooperative training time and Civic Education et al.

s.no	Unit competency	On school training		Cooperative training	Total hours	Remarks
		Theory	Practical			
1.	Provide Motivated Competent and Compassionate service	40	-	-	40	
2.	Apply Infection Prevention Techniques and Workplace OHS	32	24	40	96	
3.	Provide First Aid and Emergency Response	60	60	-	120	
4.	Perform equipment handling and maintenance	60	60	60	180	
5.	Prepare Laboratory Solutions	60	60	60	180	
6.	Collect and Process Medical Samples	100	120	80	300	
7.	Apply Computer and Mobile Health Technology	44	56	-	100	
8.	Perform Parasitological Examination	120	70	60	250	
9.	Perform Urine and Body Fluid analysis	120	70	50	240	
10.	Apply basic health statistics and health survey	40	-	40	80	
11.	Perform Community Mobilization and Provide Health Education	60	-	60	120	
12.	Apply 5S Procedures	16	8	8	32	
	Total Hours	**752**	**528**	**458**	**1738**	

1.4. Qualification Level and Certification

Based on the descriptors elaborated on the Ethiopian National TVET Qualification Framework (NTQF) the qualification of this specific TVET Program is **"**Level III".

The learner will not be awarded any certificate before completion of all the modules that are designed for the exit in level IV.

1.5. Target Groups

Any citizen who meets the entry requirements set by the concerned organization for the academic year and capable of participating in the learning activities is entitled to take part in the Program.

1.7 Entry Requirements

The prospective participants of this program are required to possess the requirements or directive of the Ministry of labor and skill.

1.8 Mode of Delivery

This TVET-Program is characterized as a formal Program on middle level technical skills. The mode of delivery is co-operative training. The TVET-institution and identified companies have forged an agreement to co-operate with regard to implementation of this program. The time spent by the trainees in the industry will give them enough exposure to the actual world of work and enable them to get hands-on experience.

The co-operative approach will be supported with school-based lecture-discussion, simulation and actual practice. These modalities will be utilized before the trainees are exposed to the industry environment.

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1.9. TVET Program Structure

Unit of Competence	Module Code & Title		Learning Outcomes	Duration (In Hours)
Provide Motivated Competent and Compassionate service	HLT MLT3 M01 0222	Providing Motivated Competent and Compassionate service	• Apply professionalism and ethical practice principles • Apply humanistic care to clients • Demonstrate effective health care communication • Provide respectful care for clients • Perform with legal and ethical framework through responsibility and accountability • Provide quality service	40
Apply Infection Prevention Techniques and Workplace OHS	HLT MLT3 M02 0222	Applying Infection Prevention Techniques and Workplace OHS	• Apply infection prevention techniques • Establish and maintain participative arrangements • Assess and control risks and hazards • Limit contamination • Clean environmental surfaces	96

Provide First Aid and Emergency Response	HLT MLT3 M03 0124	Providing First Aid and Emergency Response	• Assess and identify client's condition. • Provide first aid service • Prepare, evaluate and act in an emergency • Communicate details of the incident • Refer client requiring further care • Evaluate own performance	120
Perform equipment handling and maintenance	HLT MLT3 M04 1224	Equipment handling and maintenance	• Laboratory equipment • Equipment Operation and Handling • Equipment calibration and maintenance	180
Prepare Laboratory Solutions	HLT MLT3 05 1224	Preparing Laboratory solutions.	• Introduction. • Chemicals and Solution. • Materials used to preparing solution • Monitoring the quality of solutions • Maintain safe work environment	140
Collect and Process Medical Samples	HLT MLT3 M06 0222	Collecting and Processing Medical Samples	• Overview of anatomy and physiology • Collecting ,Handling and Transporting Medical Sample	300

			• Receiving, Distributing and logging sample • sample preparation • safe work environment	
Apply Computer and Mobile Health Technology	HLT MLT3 M07 0222	Applying Computer and Mobile Health Technology	• Start computer, system information and features • Navigate and manipulate desktop environment problems • Identify the existing Health technologies • Apply the functions of Techniques • Evaluate new or upgraded Techniques performance	100
Perform Parasitological Examination	HLT MLT4 M08 1224	Parasitological Tests	• concept of human parasitology • Processing samples and associated request details • Microscope Set up and uses • Parasitological tests • Maintaining laboratory records • safe working environment	250

Perform Urine and Body Fluid analysis	HLTMLT3 M09 1224	Urine and Body Fluid analysis	• Urinary system and concept of urinalysis • Process of urine formation • Urine test • Body fluid analysis • Safe Working environment • Laboratory records	200
Apply basic health statistics and health survey	HLT MLT3 M10 1224	Applying basic health statistics and health survey	• Prepare for the application of health survey • Undertake data collection • Compile, interpret and utilize health data • Prepare and submit reports • Take intervention measures accordingly	80
Perform Community Mobilization and Provide Health Education	HLT MLT3 M11 1224	Performing Community Mobilization and Provide Health Education	• Conduct health education and communication • Rain model families • Plan and Undertake advocacy on identified health issues	120

Apply 5S Procedures	HLT MLT3 M12 1224	Applying 5S Procedures	• Develop understanding of quality system • Sort needed items from unneeded • Set workplace in order • Shine work area • Standardize activities • Sustain 5S system	32

*The time duration (Hours) indicated for the module should include all activities in and out of the TVT institution.

1.10 Institutional Assessment

Two types of evaluation will be used in determining the extent to which learning outcomes are achieved. The specific learning outcomes are stated in the modules. In assessing them, verifiable and observable indicators and standards shall be used.

The ***formative assessment*** is incorporated in the learning modules and form part of the learning process. Formative evaluation provides the trainee with feedback regarding success or failure in attaining learning outcomes. It identifies the specific learning errors that need to be corrected, and provides reinforcement for successful performance as well. For the teacher, formative evaluation provides information for making instruction and remedial work more effective.

Summative Evaluation the other form of evaluation is given when all the modules in the program have been accomplished. It determines the extent to which competence have been achieved. And, the result of this assessment decision shall be expressed in the term 'competent or not yet competent'.

Techniques or tools for obtaining information about trainees' achievement include oral or written test, demonstration and on-site observation.

1.11 TVET Teachers Profile

The teachers conducting this particular TVET Program are A Level and have satisfactory practical experiences or equivalent qualifications.

2. Learning module design

Module Code andTitle	**HLT MLT3 M01 1224 Motivated, Competent and Compassionate service**
Nominal Duration:	**40**Hours

MODULE DESCRIPTION : This module covers the necessary knowledge, skills and attitude required to effectively form professional duties and responsibilities with compassionate caring and respectful manner by applying basic principles of professional, ethical and legal aspects of the profession.

LEARNING OUTCOMES

At the end of the module the learner will be able to:

- Apply professionalism and ethical practice principles
- Apply humanistic care to clients
- Demonstrate effective health care communication
- Provide respectful care for clients
- Function with legal and ethical framework through responsibility and accountability
- Provide quality service

MODULE CONTENTS:

Unit one: Professionalism and ethical practice

1.1. Basic concept of ethics
1.2. Professional ethics
1.3. Ethical principles
1.4. Professional code of conducts
1.5. Professional values

Unit two: Applying humanistic care to clients.

2.1 Understanding and implement clients' concern
2.2 Considering clients feelings and emotions
2.3 Defining characteristics of innate needs

Unit three: Demonstrating effective health care communication

3.1 Establishing collaborative working relationship
3.2 Expressing compassion concern of the clients
3.3 Gathering Proper information

3.4 Therapeutic communication

3.5 Non-violent communication techniques

3.6 Educating and counseling clients

Unit four: Providing respectful care for clients

4.1 Listening Health care practitioners to Clients and family perspectives choices honor

4.2 Planning and delivery of care Patient and family incorporate into knowledge, values, beliefs and cultural backgrounds

4.3 Communicating Complete and unbiased information share with patients and families

4.4 Participating Patients and families in care and decision-making

4.5 Collaborating all concerned body in policy and program development, implementing and evaluating

4.6 Respecting Patient's rights

Unit five: Performing legal and ethical framework

5.1 . Understanding Legislation and common laws

5.2 . Respecting and practice Policies

5.3 . Ensuring Confidentiality of individual's record

5.4 . Preventing patient's information Disclosure

5.5 . Ethical issues and ethical dilemma

5.6 . Handling client who is not able to communicate

5.7 . Releasing Patient-specific data

5.8 . Conducting Training programs on ethical issue for health care provider and other staff

Unit six: Providing quality service

6.1 Giving Motivated and quality service

6.2 Assisting client to gain personal goal.

6.3 Performing Task

LEARNING METHODS:

- Interactive lecture
- Group discussion
- Demonstration
- Group assignment
- Individual assignment

ASSESSMENT METHODS

- Continuous assessments (quiz, assignment, tests etc.).
- Demonstration
- Oral questioning
- Final written exam

ASSESSMENT CRITERIA

Unit one: Applying professionalism and ethical practice principles

- Identify and execute ethical principles and issues of the profession
- Identify and execute Professional code of conducts
- Recognize and demonstrate professional values
- Maintain and evaluate adherence to ethical principles of the profession
- Maintain professional practice according to applicable standards

Unit two: Applying humanistic care to clients.

- Understand and implement patients' concern
- Consider patient and clients' feelings and emotions
- Address and communicate Patients innate needs

Unit three: Demonstrating effective health care communication

- Establish positive, respectful and collaborative working relationship
- Recognize, anticipate and express compassion concern for the patient
- Gather and effectively elicit proper information in order to facilitate accurate diagnosis and management
- Use appropriate non-verbal communication
- Inform, educate and counsel clients effectively
- Establish effective interaction with other people working within the health system
- Provide therapeutic instructions compassionately
- Use and implement non-violent communication techniques

Unit four: Providing respectful care for clients

- List health care practitioners, patient and family perspectives and choices honored
- Incorporate patient and family knowledge, values, beliefs and cultural backgrounds into the planning and delivery of care
- Communicate and share complete and unbiased information with patients and families by the practitioner in in an affirming and useful manner
- Make patients and families receive timely, complete, and accurate information in order to effectively participate in care and decision-making.
- Collaborate patients, families, health care practitioners, and hospital leaders in policy and program development, implementation, and evaluation; in health care facility design; and professional education and the delivery of care.
- Respect patient's rights to access care, transfer and continuity of care

Unit five: Performing legal and ethical framework

- Understood Legislation and common laws relevant to work role
- Respect and practice policies and procedures
- Ensure confidentiality of individual's record
- Prevent disclosure of patient's information to another person without patient's consent.
- Recognize ethical issues and ethical dilemma in the workplace
- Handle patients who are not able to communicate in case of emergency or other conditions
- Release patient-specific data to only authorized users.
- Conduct training programs for health care providers and other staff on privacy and confidentiality of patient information
- Recognize and report unethical conduct

Unit six: Providing motivated and competent service

- Give motivated and qualified service for the patient
- Assist the patient to gain their personal goal.
- Perform task according to the standard.

Module Code and Title	HLT PHS3 M02 1224 First Aid and Emergency Response

Nominal Duration:	120 Hours

MODULE DESCRIPTION : This module aims to provide trainees with knowledge, skills and attitude required to recognize and respond to life threatening emergencies using basic life support, provide first aid response, management of casualty(s), the incident and other first aiders, until the arrival of medical or other assistance.

LEARNING OUTCOMES

At the end of the module the learner will be able to:

- Assess and identify client's condition
- Provide first aid service
- Prepare, evaluate and act in an emergency
- Communicate details of the incident
- Refer client requiring further care
- Evaluate own performance

MODULE CONTENTS:

UNIT ONE: CLIENT'S CONDITION ASSESSMENT

1.1. Basic principles of first aid are addressed
1.2. Hazards in the situation that may pose a risk of injury or illness
1.3. Immediate risk to self and casualty's health and safety
1.4. Selecting, using, maintaining and storing safety equipment in emergency
1.5. Vital signs and state of consciousness
1.6. History of the event is obtained.
1.7. Options for action in cases of emergency
1.8. Organizational emergency procedures and policies
1.9. Occupational health and safety procedures and safe working practices

UNIT TWO: FIRST AID SERVICE

2.1 Physical hazards controlling
2.2 Basic rules of ABC life

UNIT THREE: ACTING IN AN EMERGENCY

3.1. Options for action in cases of emergency

3.2. Grouping emergency control strategies

3.3. Occupational health and safety policies, procedures

3.4. Removing clients and other individuals from danger

3.5. Potential hazards evaluation and documentation

UNIT FOUR:COMMUNICATION DURING INCIDENT

4.1. Causality level of consciousness and communication style

4.2. first aid assistance and ambulance

4.3. Casualty's condition and management activities

4.4. Casualty's physical condition assessment and management

4.5. Confidentiality of records and information

UNIT FIVE: CLIENT REFERRAL

5.1. Client history documentation

5.2. Referral procedures documentation

5.3. Information conveying

5.4. Client care during referral.

5.5. Confidentiality of client

UNIT SIX: EMERGENCY RESPONSE EVALUATION

6.1. Clinical expert feedback

6.2. Rescuers psychological impacts

6.3. Debriefing /evaluation

LEARNING METHODS:

- Lecture
- Demonstration
- Group discussion
- Exercise
- Individual assignment

ASSESSMENT METHODS:

- Interview
- Written test
- Demonstration / observation

ASSESSMENT CRITERIA:

UNIT ONE: CLIENT'S CONDITION ASSESSMENT

- Address basic principles of first aid
- Identify hazards and its types
- Determine risk to self and casualty's health
- Maintain safety equipment in emergency
- Measure Vital signs and state of consciousness
- Obtain history of the event.
- Search options for action in cases of emergency
- Apply Organizational emergency procedures and policies
- Practice Occupational health and safety procedures and safe working practices

UNIT TWO: FIRST AID SERVICE

- Identify and control physical hazards to self and casualty's health
- Apply basic rules of ABC life

UNIT THREE: ACTING IN AN EMERGENCY

- Follow Options for action in cases of emergency
- Group emergency control strategies
- Identify Occupational health and safety policies, procedures
- Remove clients and other individuals from danger
- Assess, evaluate, report and document potential hazards

UNIT FOUR:COMMUNICATION DURING INCIDENT

- Identify causality level of consciousness and communication style
- Sough first aid assistance and ambulance support from others in a timely manner
- Observe casualty's condition and management activities
- Describe details of casualty's physical condition and management
- Keep Confidentiality of records and information

UNIT FIVE: CLIENT REFERRAL

- Documenting relevant client history
- Ensuring referral procedures documentation
- Conveying appropriate information to individuals involved in referral

- Maintaining client care during referral.
- Maintaining client confidentiality at all times and levels.
- Client confidentiality is maintained at all times and levels

UNIT SIX: EMERGENCY RESPONSE EVALUATION

- Seek feedback from appropriate clinical expert
- Address possible psychological impacts on rescuers
- Participate in debriefing/evaluation to improve future response

Module Code andTitle	HLT PHS3 M03 1224Infection Prevention Techniques and Workplace OHS
Nominal Duration:	96Hours

MODULE DESCRIPTION : This module describes knowledge, skills and attitude required for workers to comply with infection control policies and procedures. All procedures must be carried out in accordance with current infection control guidelines to ensure the workplace is safe and without risks to the health of employees, clients and/or visitors.

LEARNING OUTCOMES

At the end of the module the learner will be able to:

- Apply infection prevention techniques
- Establish and maintain participative arrangements
- Assess and control risks and hazards
- Limit contamination
- Clean environmental surfaces

MODULE CONTENTS:

Unit One: Applying infection prevention techniques

1.1. Basic components of disease transmission
1.2. Essential elements of infection prevention
1.3. Applying universal and standard precaution
1.4. Demonstrating additional precautions measures
1.5. Minimizing Contamination of materials, equipment and instruments
1.6. Performing instrumental processing
1.7. Disposing infectious/hazardous waste materials
1.8. Personal protective equipment (PPE)
1.9. Applying proper hand washing techniques

Unit Two: Establish and maintain participative arrangements

2.1 Participative processes in accordance with OHS legislation, regulations and standards.
2.2 Dealing with Issues of participation and consultation
2.3 Providing Information about the outcomes of participation and consultation

2.4 Establishing and monitoring Systems for keeping OHS records

Unit Three: Assess and control risks and hazard

3.1. Developing organizational procedures for hazard identification, assessment and control of risks.
3.2. Identification of all hazards at the planning, design and evaluation stages
3.3. Developing and maintaining hazard risk control measures
3.4. Identifying inadequacies in existing risk control measures
3.5. Protocols for care following exposure to blood or other body fluids

Unit Four: Limit contamination

4.1 Demarcating and maintaining clean and contaminated zones
4.2 Keeping records, materials and medicaments in a clean zone
4.3 Confining contaminated instruments and equipment in a well designated zone

Unit Five: Clean environmental surfaces

5.1 Wearing Personal protective equipment (PPE)
5.2 Removing dust, dirt and physical debris from worksurfaces
5.3 Cleaning work surfaces
5.4 Drying work surfaces
5.5 Replacing surface covers where applicable
5.6 Maintaining and storing equipment

LEARNING METHODS:

- Lecture-discussion
- Demonstration
- Exercise
- Individual assignment

ASSESSMENT METHODS:

- Interview
- Written test
- Demonstration/Observation

ASSESSMENT CRITERIA:

Unit One: Applying infection prevention techniques

- Identify basic components of disease transmission
- Identify Essential elements of infection prevention

- Apply universal and standard precaution
- Demonstrate additional precautions measures
- Minimize Contamination of materials, equipment and instruments
- Perform instrumental processing
- Dispose infectious/hazardous waste materials
- Identify Personal protective equipment (PPE)
- Apply proper hand washing techniques

Unit Two: Establish and maintain participative arrangements

- Ensure Participative processes in accordance with OHS legislation, regulations and standards.
- Deal with Issues of participation and consultation
- Provide Information about the outcomes of participation and consultation
- Establish and monitoring Systems for keeping OHS records

Unit Thee: Assess and control risks and hazards

- Developing organizational procedures for hazard identification, assessment and
- control of risks.
- Identification of all hazards at the planning, design and evaluation stages
- Developing and maintaining hazard risk control measures
- Identifying inadequacies in existing risk control measures
- Protocols for care following exposure to blood or other body fluids

Unit Four: Limit contamination

- Demarcate and maintain clean and contaminated zones
- Keep records, materials and medicaments in a clean zone
- Confining contaminated instruments and equipment in a well designated zone

Unit Five: Clean environmental surfaces

- Weare Personal protective equipment (PPE)
- Remove dust, dirt and physical debris from worksurfaces
- Clean work surfaces
- Drye work surfaces
- Replace surface covers where applicable

• Maintain and storing equipment

Module Code and Title	HLT PHS3 M02 1224 First Aid and Emergency Response
Nominal Duration:	120 Hours

MODULE DESCRIPTION : This module aims to provide trainees with knowledge, skills and attitude required to recognize and respond to life threatening emergencies using basic life support, provide first aid response, management of casualty(s), the incident and other first aiders, until the arrival of medical or other assistance.

LEARNING OUTCOMES

At the end of the module the learner will be able to:

- Assess and identify client's condition
- Provide first aid service
- Prepare, evaluate and act in an emergency
- Communicate details of the incident
- Refer client requiring further care
- Evaluate own performance

MODULE CONTENTS:

UNIT ONE: CLIENT'S CONDITION ASSESSMENT

1.10. Basic principles of first aid are addressed
1.11. Hazards in the situation that may pose a risk of injury or illness
1.12. Immediate risk to self and casualty's health and safety
1.13. Selecting, using, maintaining and storing safety equipment in emergency
1.14. Vital signs and state of consciousness
1.15. History of the event is obtained.
1.16. Options for action in cases of emergency
1.17. Organizational emergency procedures and policies
1.18. Occupational health and safety procedures and safe working practices

UNIT TWO: FIRST AID SERVICE

2.3 Physical hazards controlling
2.4 Basic rules of ABC life

UNIT THREE: ACTING IN AN EMERGENCY

3.6. Options for action in cases of emergency

3.7. Grouping emergency control strategies
3.8. Occupational health and safety policies, procedures
3.9. Removing clients and other individuals from danger
3.10. Potential hazards evaluation and documentation

UNIT FOUR:COMMUNICATION DURING INCIDENT

4.6. Causality level of consciousness and communication style
4.7. first aid assistance and ambulance
4.8. Casualty's condition and management activities
4.9. Casualty’s physical condition assessment and management
4.10. Confidentiality of records and information

UNIT FIVE: CLIENT REFERRAL

5.6. Client history documentation
5.7. Referral procedures documentation
5.8. Information conveying
5.9. Client care during referral.
5.10. Confidentiality of client

UNIT SIX: EMERGENCY RESPONSE EVALUATION

6.4. Clinical expert feedback
6.5. Rescuers psychological impacts
6.6. Debriefing /evaluation

LEARNING METHODS:

- Lecture
- Demonstration
- Group discussion
- Exercise
- Individual assignment

ASSESSMENT METHODS:

- Interview
- Written test
- Demonstration / observation

ASSESSMENT CRITERIA:

UNIT ONE: CLIENT'S CONDITION ASSESSMENT

- Address basic principles of first aid
- Identify hazards and its types
- Determine risk to self and casualty's health
- Maintain safety equipment in emergency
- Measure Vital signs and state of consciousness
- Obtain history of the event.
- Search options for action in cases of emergency
- Apply Organizational emergency procedures and policies
- Practice Occupational health and safety procedures and safe working practices

UNIT TWO: FIRST AID SERVICE

- Identify and control physical hazards to self and casualty's health
- Apply basic rules of ABC life

UNIT THREE: ACTING IN AN EMERGENCY

- Follow Options for action in cases of emergency
- Group emergency control strategies
- Identify Occupational health and safety policies, procedures
- Remove clients and other individuals from danger
- Assess, evaluate, report and document potential hazards

UNIT FOUR:COMMUNICATION DURING INCIDENT

- Identify causality level of consciousness and communication style
- Sough first aid assistance and ambulance support from others in a timely manner
- Observe casualty's condition and management activities
- Describe details of casualty's physical condition and management
- Keep Confidentiality of records and information

UNIT FIVE: CLIENT REFERRAL

- Documenting relevant client history
- Ensuring referral procedures documentation
- Conveying appropriate information to individuals involved in referral

- Maintaining client care during referral.
- Maintaining client confidentiality at all times and levels.
- Client confidentiality is maintained at all times and levels

UNIT SIX: EMERGENCY RESPONSE EVALUATION

- Seek feedback from appropriate clinical expert
- Address possible psychological impacts on rescuers
- Participate in debriefing/evaluation to improve future response

Module Code and Title	HLT MLT3 M04 1224 Equipment handling and maintenance
Nominal Duration:	**180** Hours

MODULE DESCRIPTION : This module covers knowledge, skills and attitude required to equipment handling and maintenance procedure. Identifying different parts of equipments, implementing equipment operation, performing calibration and maintenance using standard calibration procedures. These procedures specify all associated reference standards, materials, equipment and methods to be used and the required parameters or quantities and ranges to be tested, including the criteria for rejection or approval

LEARNING OUTCOMES

At the end of the module the learner will be able to:

- Identify different parts of laboratory equipment
- . Perform equipment Operation and Handling
- Perform equipment calibration
- Perform equipment maintenance
- Keep records of equipment maintenance and calibration report

Module Contents:

Unit-One: laboratory equipment

1.1 parts of laboratory equipments

1.2 Function of laboratory equipments

1.3 Use of laboratory equipments.

1.4 instructions or modifications Of methods

Unit-Two: Equipment Operation and Handling

2.1 Monitoring environmental conditions of equipments

2.2 Occupational health and safety

2.3 Principle of Laboratory equipment.

2.4 Equipment operation

2.5 Equipment Safety precaution and handling

Unit-Three : Equipment calibration and maintenance

3.1. Requirements of calibration.

3,2. Selecting authorized calibration procedures

3.3 Reference standards and assembling

3.4 Verification of reference standards and measuring equipment

3.5 Adjusting or calibrating equipments

3.6 validation and recording of readings

3.7 Analyzing generated calibration report

3.8 Listing calibration approval and rejection requirements.

3.9 Arranging internal peer checking of calibration procedure

3.10Comparing test results with other laboratories.

3.11Confirming calibration procedures are fit for purpose.

3.12Looking for appropriate advice when interpretation for unusual results

Learning Methods:

- Interactive lecture
- Group discussion
- Demonstration
- Group assignment
- Individual assignment

Assessment Methods:

- Continuous assessments
- Oral questioning
- Skill check out
- Final written exam

Assessment Criteria

Unit-One: laboratory equipment

- Describe parts of laboratory equipments
- Describe the function of laboratory equipments
- Ensure appropriate use of laboratory equipments.
- Describe new instructions or modifications to methods

Unit Two Equipment Operation and Handling

- Monitor environmental conditions of equipments
- Utilize appropriate PPE
- Identify the principle of each equipment.
- Implement equipment operation
- Consider equipment Safety precaution and handling

Unit Three Equipment calibration and maintenance

3.1 Confirm requirements of equipments for calibration procedure.
3.2 Select authorized calibration procedures
3.3 Specify reference standards and assembling associated equipments
3.4 Verify the performance of reference standards and measuring equipment
3.5 Adjust or calibrate equipments
3.6 Confirm the reading of the measuring results are valid and record
3.7 Analyze generated calibration report to detect trends or inconsistencies.
3.8 List calibration approval and rejection requirements.
3.9 Arrange internal peer checking of calibration procedure, data and results
3.10 Compare test results with other laboratories.
3.11 Confirm calibration procedures are fit for purpose.
3.12 Look for appropriate advice when interpretation for unusual results

Module Code and Title	HLT MLT3 06 0621 Preparing Laboratory solutions.
Nominal Duration:	**140** Hours.

Module Description: This unit of competency covers the ability to choose reagent grades, determine desired quantity, perform required dilution, prepare solution, standardize solution, and monitor the quality of solutions and its storage condition.

TRAINEE'S OUTCOMES

At the end of the module the learner will be able to:

- Prepare laboratory solutions following SOPs.
- Standardize solution.
- Monitor quality of laboratory solutions.
- Maintain safe work environment

Module Contents:

MODULE CONTENTS:

Unit-1: Prepare a working solutions

1.1. Introduction

1.2. Define terminologies.

1.3. Measurement.

1.3.1. Measurement units.

1.3.1.1. Physical units

1.3.1.2. Chemical units

1.3.2. Estimate uncertainty of measurement.

1.3.3. Accuracy and precision.

Unit-2: Chemicals and Solutions

2.1. Introduction.

2.1.1 Chemicals.

2.1.1.1 Definition of chemicals.

2.1.1.2 Characteristics of chemicals.

2.1.1.3 Grade of chemicals.

2.1.2 Solutions.

2.1.2.1 Definition of solution.

2.1.2.2 Types of solutions.

2.1.2.3 Expressing concentration of solutions.

2.1.2.3.1. Relative Expression

2.1.2.3.2. Quantitative Expressions of Concentration

2.1.3 Chemistry of acids, base, buffer and chemical reaction.

2.1.4 Making dilution.

2.1.4.1 Simple dilution.

2.1.4.2 Serial dilution.

2.2 Standard solutions.

2.2.1 Types of standard solution.

2.2.2 Determine the concentration of standardize solutions.

2.2.2.1 Titration.

2.2.2.2 Others methods.

Unit-3: Materials used to preparing solution

3.1. Laboratory glass wares and plastic wares

3.2. Classification of Laboratory glass wares

3.2.1. Pipettes

3.2.2. Burettes

3.2.3. Flasks

3.2.4. Beakers

3.2.5. Cylinders

3.2.6. Test tube

3.2.7. Reagent bottles

3.2.8. Reagent bottles

3.2.9. Pestle and mortar

3.2.10. Pasture pipette

3.3. Equipment for purifying water

3.4. Equipment for weighing/Balances

3.5. Incubetor

3.6. Mixer

Unit-4: Preparing laboratory solutions.

4.1. Types of clinical laboratory solution

4.1.1. Reagents solutions
4.1.2. Stock reagent solution
4.1.3. Working –reagent solution
4.2. Staining solution
4.2.1. Basic Stains
4.2.2. Acidic Stains
4.2.3. Neutral Stains
4.3. Calculating and recording data
4.4. Measuring chemicals for solution preparation
4.5. Preparation of working solution
4.6. Labeling and storage of reagents
4.7. Record working solution details in laboratory register
4.8. Cleaning and storage of materials and equipment
4.9. Errors in the preparation of solutions
4.10. Standardization of solutions.

Unit-5: Monitoring the quality of solutions

5.1. Check the quality of prepared solution before use.
5.2. Monitor the quality of stored solution.
5.3. Record quality monitoring details.

Unit-6: Maintain safe work environment

4.1. Safe use of laboratory chemicals, glassware and equipment.
4.2. Follow safe work practices.
4.3. Store equipment and reagents as required.

Learning Methods:

- Interactive lecture.
- Group discussion.
- Demonstration.
- Role playing.
- Practice with exercise.
- Problem based learning Demonstration.
- Apprenticeship.

Assessment Methods:

- Continuous assessments (quiz, assignment, tests etc.).
- Oral questioning (Interview).
- Presentation.
- Practical demonstration.
- Practical assessment by direct observation of tasks using prepared checklist.
- Objective based clinical examination.
- Final written exam.

Assessment Criteria

Unit-1: Introduction

- Terminologies are defined appropriately.
- Specifying measurement physical and chemical units appropriately.
- Accurate and precise measurement of data.
- Estimate uncertainty of measurement accurately and precisely.

Unit-2: Chemicals and Solutions

- The relevant/appropriate standard procedure is selected for stock solution and/or working solutions preparation.
- Materials and solvent of specified purity are selected.
- Data is calculated and recorded.
- Appropriate quantities of reagents are measured for solution preparation and record data.
- Specified laboratory Equipment and appropriate grade of glassware are selected and assembled.
- The required working solution is mixed or diluted in accordance with procedures.
- Solutions are prepared to achieve homogeneous mix of the specified concentration.
- Solutions are labeled and stored to maintain identity and stability.
- Working solution details are recorded in laboratory register.
- Appropriate laboratory equipment are assembled.
- Serial dilutions are performed, as required.
- The solution to the required specified range and precision is standardized.
- The concentration of standardize solutions is determined.
- Solutions are labeled and stored to maintain identity and stability re-standardized if require.

Unit-3: Monitoring the quality of solutions

- The quality of prepared solution is checked before use.
- The quality of stored solution is monitored.
- Quality monitoring details are recorded.

Unit-4: Maintain safe work environment

- Appropriate safety precautions are applied for use of laboratory equipment and hazardous chemical materials.
- Appropriate laboratory glassware and measuring equipment are used.
- Established safe work practices and PPE are used to ensure personal safety and that of other laboratory personnel.
- Spills are cleaned up using appropriate techniques to protect personnel, work area and environment.
- Generation of waste and environmental impacts are minimized.
- The safe collection of laboratory hazardous waste for subsequent disposal is ensured.
- Glassware and equipment are cleaned and stored in accordance with enterprise procedures.
- Equipment and reagents are stored as required.

MODULE CODE AND TITLE:	HLT MLT3 M06 0424 Collecting and Processing Medical Samples
NOMINAL DURATION:	300 Hours

MODULE DESCRIPTION: This module covers knowledge, skills and attitude required to apply concepts of physiology and anatomy of human, collect, handle, and transport and prepare samples for testing at work site or field using specified equipment and standard or routine procedures in a way that ensures the integrity of subsequent samples.

Training Outcomes

At the end of the module the learner will be able to:

- Describe concepts of anatomy and physiology
- Collect ,Handle and Transport Medical Sample
- Receive, Distribute and log sample
- Prepare sample for testing.
- Maintain safe work environment

MODULE CONTENTS:

Unit one: Overview of anatomy and physiology

1.1 Basics of human anatomy and physiology

1.2 Type and nature of samples

1.3 Samples collection time and sites

Unit two: Collecting ,Handling and Transporting Medical Sample

2.1 Collection ,handling and transporting stool sample

2.2 Collection ,handling and transporting Urine sample

2.3 Collection ,handling and transporting Blood sample

2.4 Collection ,handling and transporting sputum sample

2.5 Collection ,handling and transporting Cerebrospinal Fluid(CSF) sample

2.6 Collection ,handling and transporting discharge sample

2.7 Collection ,handling and transporting Skin sample

Unit 3 :- Receiving, Distributing and logging sample

- Sample acceptance and rejection criteria
- Laboratory Information Management System

Unit 4: - Preparing sample for testing.

- Physical and Chemical separation of the samples
- Samples storing

Unit 5:- safe work environment

- Establishing work practices and using PPE
- Cleaning equipment, containers, work area and vehicles
- Laboratory waste management

LEARNING METHODS:

- Lecture
- Demonstration
- Simulation
- Role-play
- Group discussion
- Exercise
- Individual assignment

ASSESSMENT METHODS:

- Written exam/test
- Practical assessment
- Questioning or interview
- Oral examination

ASSESSMENT CRITERIA:

Unit one: Overview of anatomy and physiology

- Describe basics of human anatomy and physiology

- Describe type and nature of samples
- Identify time of samples collection and collection sites

Unit two: Collecting ,Handling and Transporting Medical Sample

- Perform Collection ,handling and transporting stool sample
- Perform Collection ,handling and transporting Urine sample
- Perform Collection ,handling and transporting Blood sample
- Perform Collection ,handling and transporting sputum sample
- Perform Collection ,handling and transporting Cerebrospinal Fluid(CSF) sample
- Perform Collection ,handling and transporting discharge sample
- Perform Collection ,handling and transporting Pus sample
- Perform Collection ,handling and transporting Skin sample

Unit 3 :- Receive, Distribute and log sample

- Describe Sample acceptance and rejection criteria
- Perform Laboratory Information Management System

Unit 4: - Prepare sample for testing.

a. Perform Physical and Chemical separation of the samples
b. Apply Samples stored in accordance with SOPs.

Unit 5:- Maintain safe work environment

- Establishing work practices and using PPE
- Cleaning equipment, containers, work area and vehicles
- Perform Laboratory wastes managemnt

Module Code and Title	HLT PHS3 M08 1224 Basic Computer and mobile health technology
Nominal Duration:	**100**Hours

MODULE DESCRIPTION : This module aim to covers the necessary knowledge, skills and attitude required to store, retrieve, analyze, present and file health data using word processing, spreadsheet, access and power point applications. It also includes the competence required to identify, utilize internet services and use for new or upgraded technology.

LEARNING OUTCOMES

At the end of the module the learner will be able to:

- Connect computer hardware peripherals
- Install and administer basic computer applications
- Utilize software applications
- Utilize internet services
- Identify the existing Health Technologies
- Apply the functions of technology

MODULE CONTENTS:

Unit one: Connecting computer hardware peripherals

1.1 Adjusting workspace, furniture and equipment in line with ergonomic requirements.
1.2 Identifying computer components and types.
1.3 Installing computer components and setting ready to use.
1.4 Connecting printer with computer and setting ready to use.

Unit two: Installing and administering basic computer applications

2.1 Installing basic computer applications.
2.2 Installing required device drivers.
2.3 Creating and administering user accounts

Unit Three: Utilizing software applications

3.1 Identifying appropriate software application for data processing
3.2 Handling data process with Ms Office word processor.
3.3 Demonstrating data analysis and processing with Ms excel.
3.4 Creating database for health data management and used with Ms Access.
3.5 Data presentation with Mspower point.

3.6 Designing activities with Ms Visio.

3.7 Handling work related data with notepad or word pad.

Unit Four: Utilizing internet services

4.1 Internet/intranet services

4.1.1. Introduction to internet service

4.1.2. Internet, access services and its application

4.2 Identifying resources and retrieving on the internet.

4.3 Using email service for report and feedback.

4.4 Identifying and using web-based e-health applications.

Unit Five: Identify the existing Health Technologies

5.1 Identifying the existing knowledge and techniques to technology.

5.2 Mobile technology skills to enhance learning and provision of standard health care

5.3 Applying M-health techniques to enhance efficient utilization of resources

5.4 Identifying and using new and updated equipment's

Unit six: Applying the functions of technology

6.1 Computer applications (technology)

6.2 Testing of new or upgraded equipment

6.3 Features of new or upgraded equipment.

6.4 Accessing, interpreting and using source of information to upgraded equipment

LEARNING METHODS:

- Interactive lecture
- Demonstration
- Group discussion
- Group/Individual assignment

ASSESSMENT METHODS:

- Continuous assessments (quiz, assignment, tests etc.).
- Oral questioning
- Skill check-out (Demonstration/Observation)
- Final written exam

ASSESSMENT CRITERIA:

Unit One: Connecting computer hardware peripherals

- Adjust workspace, furniture and equipment to suit user ergonomic requirements.
- Identify computer components and types.
- Install and set computer components ready to use.
- Set printer connected with computer and ready for use.

Unit two: Installing and administering basic computer applications

- Install basic computer applications.
- Install all required device drivers.
- Create and administer user account.

Unit three: Utilizing software applications

- Install appropriate software application for data processing
- Process data with Microsoft office word processor.
- Demonstrate data analysis and process with Microsoft excel.
- Create and use database for health data management with Microsoft Access.
- Work out data presentation with MS power point presentation.
- Work out design activities with MS Visio.
- Handle work related data with Notepad or word pad.

Unit four: Utilizing internet services

- Identify and execute Internet/intranet services.
- Search, identify and retrieve resources on the internet
- Send reports to the next level using e-mail services and received feedback.
- Identify and use Web based e-health applications.

Unit five: Identifying the existing Health Technologies

- Identify the existing knowledge and techniques to technology.
- Apply mobile technology skills to enhance learning and provision of standard health care.
- Apply M-health techniques to enhance efficient utilization of resources.
- Identify and use upcoming and updated equipment's

Unit six: Applying the functions of technology

- Apply computer applications (technology) for Health data collection, organization, analysis and interpretation.
- Test new or upgrade equipment according to the specification manual.
- Identify features of new or upgraded equipment.

- Apply access, interpret and use source of information to upgraded equipment.

Module Code and Title	HLT MLT3 M08 0222 Parasitological Tests
Nominal Duration:	**250** Hours

MODULE DESCRIPTION : This module covers knowledge, skills and attitude required to detect and differentiate the structure and stage of parasites using basic tests and procedures identified with the discipline of parasitological laboratory using microscope and other methods

LEARNING OUTCOMES

At. the end of the module the learner will be able to:

- Describe concept of human parasitology
- Process samples and associated request details
- Set up and uses Microscope
- Perform tests
- Maintain laboratory records
- Maintain a safe environment

MODULE CONTENTS:

Unit One: Human parasitology

1.1. Basic of human parasitology
1.2. principle of host and parasite interaction
1.3. life cycles and diagnostic stage of parasites
1.4. methodology of parasitological examinations
1.5. Microscope set up and uses

Unit two: Process samples and associated request details

2.1. Checking samples and request form
2.2. Sorting specimens
2.3. Rejecting samples and request forms with reasons for non-acceptance
2.4. Logging acceptable samples and requested forms

2.5. Processing samples

2.6. Storing Samples and sample components

Unit Three: Microscope Set up and uses

3.1. Setting up the light path

3.1.1. Adjusting settings and alignment

3.2. Selecting and filtering objectives

3.3. Ensuring the lenses are made clean

Unit Four: Perform tests

4.1. Select authorized tests for the requested investigations

4.2. Perform quality control procedures

4.3. Conduct Basic parasitological tests

4.4. Record results

4.5. Verify results before releasing for clinician/client

4.6. Discussing with colleague when results are panic

4.7. Storing unused sample or sample components properly.

Unit-5: Maintain woke environment

5.1. Using PPE and established safe work practices

5.2. Cleaning up spills using appropriate techniques

5.3. Minimizing generation of wastes

5.4. Ensuring safe disposal of biohazardous materials and other laboratory wastes

Unit-6. Maintain laboratory record

6.1. Entering on report forms or into computer systems.

6.2. Updating instrument maintenance logs

6.3. Maintaining Security and confidentiality

Learning Methods:

- Lecture
- Demonstration
- Group discussion
- Exercise
- Individual assignment

Assessment Methods:
• Practical assessment • Written exam/test • Questioning or interview
Assessment Criteria

Unit One: Human parasitology

- Describe basics of human parasitology
- Describe principle of host and parasite interaction
- Identify life cycles and diagnostic stage of parasites
- Demonstrate methodology of parasitological examinations
- Examine microscope set up and uses

Unit two: Process samples and associated request details

- Check samples and request form
- Sort specimens
- Reject samples and request forms with reasons for non-acceptance
- Log acceptable samples and requested forms
- Process samples
- Store Samples and sample components

Unit Three: Microscope Set up and uses

- Set up the light path
- Select and filtering objectives
- Ensure the lenses are made clean

Unit Four: Parasitological tests

- Select authorized tests for the requested investigations
- Perform quality control procedures
- Conduct Basic parasitological tests
- Record results
- Verify results before releasing for clinician/client
- Discuss with colleague when results are panic
- Store unused sample or sample components properly.

Unit-5: Maintain woke environment

- Use PPE and established safe work practices
- Clean up spills using appropriate techniques
- Minimize generation of wastes
- Ensure safe disposal of bio hazardous wastes

Unit-6. Maintain laboratory record

- Enter on report forms or into computer systems.
- Up date instrument maintenance logs
- Maintaining Security and confidentiality
- Maintaining Security and confidentiality

Module Code and title	HLTMLT3 M10 42024Urine and Body Fluid analysis
Nominal duration	250

MODULE DESCRIPTION: This module aims to provide the traineeswith the knowledge, skills and right attitudes required to determine the type and quantity of different metabolic products in urine and identification of the different components of urine sediments using tests and procedures identified with the discipline of urinalysis laboratory.

LEARNING OUTCOMES: At the end of the module the learner will be able to:

- Identify urinary system and concepts of urinalysis
- Explain Process ofurine formation
- Explain type of Urine Specimen, Collection and Preservation
- Perform Urine test
- Perform Body Fluid Analysis
- Maintain laboratory records
- Maintain a safe environment

MODULE CONTENTS:

Unite One;Urinary system and concepts of urinalysis

1.1 Anatomy and physiology of urinary system.
1.2 External Anatomy of the kidney
1.3 Internal Anatomy of the kidney
1.4 Physiology of the Kidney

Unite two: Process of urine formation

2.1 Formation of Urine
2.2 Composition of Urine
2.3 Renal Clearance
2.4 Renal Threshold:

Unite Three: Urine Specimen Collection and Preservation

3.1 Types of urine specimen

3.1.1 Early/first morning urine specimen

3.1.2 Random urine specimen

3.1.3 24-Hour urine specimen

3.1.4 Midstream urine specimen

3.1.3 Terminal urine specimen

3.1.4 Clean catch urine specimen

3.1.5 Urine specimens collected using a catheter

3.1.6 Urine specimens from infants:

3.2 Specimen acceptance and Rejection criteria

3.3Urine preservation techniques.

Unite Four: Urine test

3.3.1 Physical Examination of Urine

3.3.2 Chemical Examination of Urine

3.3.3 Microscopic Examination of Urine

Unite Five: Body Fluid Analysis

4.1 Semen Analysis

4.1.1 Measurement of volume:

4.1.2 Measurement of pH

4. 1. 3 Sperm count:

4.2 Cerebrospinal fluid (CSF)

4.2.1 Common reasons for investigation of CSF

4.2.2 Glucose test

4.2.3 Protein test

4.2.4 Gram staining

4.2.5 AFB staining

4.2.6 Differential count

4.2.7 White blood cells count

Unite Six:Recording and Reporting of Result

6.1 Recording and Standardization in reporting test results

Unite Seven:Safe Working environment

7.1 Universal (Standard) Precautions:

7.2Techniques to clean Spills

7.3Waste generation and Segregation

7.4 Safe disposal of laboratory wastes.

LEARNING METHODS:

- Interactive lecture
- Group exercises
- Role play simulations
- Case study
- Practical exercise - demo and practice with coaching
- Co-operative training

ASSESSMENT METHODS:

- Competence may be accessed through:

- Interview/Written Test
- Observation/Demonstration(Practical) with Oral Questioning

ASSESSMENT CRITERIA:

Unite one: Urinary system and concepts of urinalysis

- Discus Anatomy of urinary system.
- Identify External Anatomy of the kidney
- Identify Internal Anatomy of the kidney
- Identify Physiology of the Kidney

Unite Two: Process of urine formation

- Explain Formation of Urine
- Identify Composition of Urine

- Describe Renal Clearance
- Discus Renal Threshold:

Unite Three: Urine Specimen Collection and Preservation

- Describe types of urine specimen
- Explain specimen acceptance and Rejection criteria
- Describe urine preservation techniques.

Unite Four:Urine test

- Identify type of urine examination
- Explain the physical examination of urine
- Explain the chemical examination of urine
- Explain the microscopic examination of urine

Unite Five: Body Fluid Analysis

- Perform Measurement of Semen volume
- Perform Measurement of pH
- Perform Sperm count:
- Explain Common reasons for investigation of CSF
- Perform Glucose test
- Perform Protein test
- Perform Gram staining
- Perform AFB staining
- Perform Differential count
- Perform White blood cells count

Unite Six:Recording and Reporting of Result

- Entries data into laboratory information systemaccurately

- Maintain Records of received Urine and Body Fluid samples
- Maintain Security and confidentiality of all clinical information
- MaintainedLaboratory data and records

Unite Seven:Maintain a safe environment

- Explain Universal (Standard) Precautions:
- Identify Techniques to clean Spills
- Discus Waste generation and Segregation
- Explain Safe disposal of laboratory wastes.

LEARNING METHODS:

• Lecture

• Demonstration

• Group discussion

• Exercise

• Individual assignment

ASSESSMENT METHODS:

- Interview
- Written test
- Demonstration/Observation

ASSESSMENT CRITERIA:

LO.1 Identify different parts of laboratory equipment

- Parts of laboratory equipments are described
- Function of laboratory equipments are described
- Appropriate use of laboratory equipments is ensured.
- All new instructions or modifications to methods are described to ensure repeatability of test.

LO.2 Perform equipment Operation and Handling

- Environmental conditions are monitored to ensure proper function of equipment according to manufacturer specification.
- The appropriate PPE are used during equipment operation
- Principle of each equipment is identified.
- Equipments are properly operated according to the manufacturer recommendation.
- Safety precaution are considered during equipment operation and handling

LO.3 Perform equipment calibration

- All measuring equipment are confirmed to meet the laboratory's specification requirements and complied fully with the calibration procedure
- The authorized calibration procedures are selected in accordance with enterprise procedures
- Specified reference standards and associated equipment are assemble and set up prior to testing

- Performance of reference standards and measuring equipment is verified prior to use and adjusted or calibrated, as necessary
- Confirm readings are the result of a valid measurement and record data as required (before and after adjustment)
- Device under test is adjusted to bring readings within specification and data recorded (after and after adjustment), if required
- Generated calibration report is analyzed to detect trends or inconsistencies that would significantly affect the accuracy or validity of test results
- The requirements are listed for calibration approval and rejection.
- Internal peer checking of calibration procedure, data and results are arranged for and feedback incorporated.
- Results are compared with those obtained by other laboratories, if applicable.
- Confirm that the calibration procedure is fit for purpose and relevant to the client's needs
- Appropriate advice is sought when interpretation of results is outside authorized scope of approval

LO.4 Perform equipment maintenance

- Preventive maintenance activities are conducted according to manufacture requirement of each equipment.
- Troubleshooting is performed for identified errors according to manufacturer’s recommendation
- Equipment having errors that requires trained biomedical engineers are identified
- Knowledge and practice is applied to verify equipment maintenance by service engineers
- Potential sources of measurement error are identified and minimized

LO5 Keep records of equipment maintenance and calibration report

- The procedure is reported and presented to appropriate personnel for approval before use
- Prepare and issue a final report on the job/item detailing testing carried out, traceability, statement of compliance and relevant information as required
- Any non-compliance is reported and next course of action verified.
- Calibration labels, equipment stickers, quality control tags and tamper resistant seals are attached, as required in enterprise procedures
- Compliance/non-compliance is documented with requirements of test and/or specifications
- Test equipment/measurement standards and results are stored in accordance with enterprise procedures

Annex: Resource Requirements

HLT MLT3 M04 1224 Performing equipment handling and maintenance				
Item No.	**Category/Item**	**Description/ Specifications**	**Quantity**	**Recommended Ratio (Item: Learner)**
A.	**Learning Materials**			
1.	TTLM	• Prepared by the trainer	25	1:1
2.	Textbooks	-	5	1:5
3.	Reference Books			
3.1	District Laboratory practice in tropical countries Part I	Monica cheesbrough. second edition; 2009	5	1:5
4.	Journals/Publication/Magazines	• Equipment quality Indicators/latest • Fact sheets • Standard formats	10	1:3
4.1				
B.	**Learning Facilities & Infrastructure**			
1.	Lecture Room	5*5m	1	1:25
2	Laboratory room	5*10 m	1	1:25
3.	Library	Standard	1	
C.	**Consumable Materials**			

1.	Paper	A4	5rim	1:5
2.	Pen	Standard		
3	Pencil and rubber	Standard		
4	Graph paper	Standard		
5	Bucher paper	Standard	10	1:3
6	Marker	Standard	12 per pack	
7	Printer ink	Standard	4	
8	White board marker	Standard	15	
9	Plaster	Standard		
10	Gloves	Standard		
11	Micro tips	Standard		
12	Distilled water	Standard	1	1:25
13	Standard	solution	1	1:25
14	Calibrator	solution	1	1:25
15	Reference materials	solution	1	1:25
16	Filter Paper	Standard		
17	Pasteur pipette	Standard		
18	Lens Paper	Standard		
19	Cuvette	1cm		
20	Test tube	Standard		
D.	**Tools and Equipment**			
1.	Computer	Lap top	1	1:25
2.	LCD projector	LCD Projector	1	1:25
3.	Printer	Laser Jet	1	1:25
4	Photocopy machine	Standard	1	1:25
5	Scanner	Standard	1	1:25
6	Back up	Standard	1	1:25
7	White board	110X80mm	1	1:25
8	Microscope	Binocular & Light	5	1:5
9	Centrifuge	Macro type	2	1:15
10	Hematocrite centrifuge	Microtype	2	1:15

11	Tachometer	standard	1	1:25
12	clinical chemistry analyzer	Photometer and Semi-automated	1	1:25
13	Balance	Manual	1	1:25
14	Digital balance	Sensitive type	1	1:25
15	Micropipette	0-50 µl 50-100 µl 100-1000 µl	2 for each	1:15
16	Complete blood cell count analyzer	Three differential	1	1:25
17	Autoclave	Standard		
18	Oven	Standard		

LO3.Monitor the quality of laboratory solutions

3.1 Checking the quality of prepared solution

3.2 Monitoring the quality of stored solution

3.3 recording Quality monitoring

4. Maintain safe work environment

4.1. Applying safety precautions

4.2. Using appropriate laboratory glassware and measuring equipment

4.3. Using PPE and established safe work practices.

4.4. Minimizing generation of waste and environmental impacts

4.5. Ensuring safe collection of laboratory hazardous waste

4.6. Cleaning and storing Glassware, reagents and equipment.

LEARNING METHODS:

- Lecture
- Demonstration
- Group discussion
- Exercise
- Individual assignment

ASSESSMENT METHODS:

- Interview
- Written test
- Demonstration/Observation

ASSESSMENT CRITERIA:

LO.1 Prepare a working solution

- The relevant/appropriate standard procedure is selected for stock solution and/or working solutions preparation
- Materials and solvent of specified purity are selected
- Data is calculated and recorded
- Appropriate quantities of reagents are measured for solution preparation and record data
- Specified laboratory Equipment and appropriate grade of glassware are selected and assembled
- The required working solution is mixed or diluted in accordance with procedures
- Solutions are prepared to achieve homogeneous mix of the specified concentration
- Solutions are labeled and stored to maintain identity and stability
- Working solution details are recorded in laboratory register

LO.2 Standardize solution

- Appropriate laboratory equipment are assembled
- Serial dilutions are performed, as required
- The solution to the required specified range and precision is standardized
- The concentration of standardize solutions is determined
- Solutions are labeled and stored to maintain identity and stability re-standardized if require

LO.3 Monitor the quality of laboratory solutions

- The quality of prepared solution is checked before use
- The quality of stored solution is monitored
- Quality monitoring details are recorded

LO4 Maintain safe work environment

- Appropriate safety precautions are applied for use of laboratory equipment and hazardous chemical materials
- Appropriate laboratory glassware and measuring equipment are used
- Established safe work practices and PPE are used to ensure personal safety and that of other laboratory personnel
- Spills are cleaned up using appropriate techniques to protect personnel, work area and environment
- Generation of waste and environmental impacts are minimized
- The safe collection of laboratory hazardous waste for subsequent disposal is ensured
- Glassware and equipment are cleaned and stored in accordance with enterprise procedures
- Equipment and reagents are stored as required

Module Code and Title	**HLT PHS3 M10 1224 Health Care Statistics, Survey and HMIS indicators**
Nominal Duration:	**80**Hours

MODULE DESCRIPTION : This module aims to provide the trainees with the knowledge, skills and right attitudes required to prepare and interpret basic healthcare statistics, health survey and compute health management information system (HMIS) indicators.

LEARNING OUTCOMES

At the end of the module the learner will be able to:

- Identify variables
- Collect Survey, HMIS and related data
- Organize and Store data
- Analyze and utilize health data
- Compute HMIS indicators
- Compute Key Performance indicators
- Compute vital statistics
- Disseminate and Utilize HMIS information

MODULE CONTENTS:

Unit two: Basics of health care statistics

1.1 Introduction to health care statistics
1.2 Variables and datasets
1.3 Abbreviations, Terminologies, and data specifications.
1.4 Resources, limitations, and assumptions for dataset size.
1.5 Data categorization

Unit two: Data collection in health care service

2.1 Methods of data collection
2.2 Data collection tools
2.3 Data collection process

Unit three: Data Organization and Storage

3.1 Data organization
3.2 Medium of data storage
3.3 Structure and formats for storing and managing data

3.4 Interval of categorical data

3.5 Variables analysis

Unit four: Analysis of Health care Data

4.1. Data Summarization

4.2. Data Analysis

4.3. Data Presentation

4.4. Degrees of uncertainty and assumptions

4.5. Data Interpretation

Unit five: Computing HMIS indicators

5.1. Introduction HMIS

5.2. HMIS indicators

5.3. Defining HMIS indicators

5.4. Interpretation of HMIS indicators

Unit six: Key Performance indicators

6.1. KPIs for Hospitals and Health centers

6.2. KPI formula

6.3. KPIs interpretation

Unit Seven: Vital statistics

1.1. Basics of Vital statistics

1.2. Civil registration

1.3. Computing and interpreting vital statistics

Unit eight: HMIS information Utilization and Dissemination

8.1. Target audiences for information dissemination.

8.2. Methods for HMIS Presentation and dissemination

8.3. Organizational Information Dissemination

8.4. Receipt of output and feedback in HMIS

8.5. HMIS information use

LEARNING METHODS:

- Lecture
- Demonstration
- Group discussion
- Exercise

Methods of Assessment

- Interview/Oral question
- Written Test
- Quiz
- Demonstration, Observation
- Group or individual Assignment

Module Code andTitle	HLT PHS3 M08 1224 Community Mobilization and Provide Health Education
Nominal Duration:	120Hours

MODULE DESCRIPTION : This module aims to provide the trainees with the knowledge, skills and right attitudes attitude required to undertake health education, advocacy and community mobilization on identified health issues.

LEARNING OUTCOMES

At the end of the module the learner will be able to:

- Conduct health education and communication
- Train model families
- Plan and Undertake advocacy on identified health issues

MODULE CONTENTS:

Unit One: health education and communication

1.1. Concept of health and health education
1.2. Assessment and identify gaps
1.3. Community organization and all available resources
1.4. Target group
1.5. Health education plan
1.6. Methods and approaches of health communication
1.7. Health education service
1.8. Monitoring of service utilization and evaluation of behavior change
1.9. Internal and external dissemination of information
1.10. Maintaining work related network and relationship
1.11. Approaches to meet communication needs

Unit Two: Train model families

2.1. Better performing households
2.2. Establishment space and time for training
2.3. Collection for resources for training
2.4. Training according to MOH guidelines
2.5. Post training follow up and monitoring
2.6. Certification for well performing model households

Unit Three: Plan and Undertake advocacy on identified health issues 3.1. advocacy plan to address health issues 3.2. Consult community representatives' priorities health needs 3.3. Influential community representatives and health development armies to disseminate IEC- BCC activities 3.4. Implementing of advocacy and community mobilization 3.5. Organizing continuous advocacy services in partnership with stakeholders 3.6. Utilizing feedback for planning
LEARNING METHODS: • Lecture and discussion • Role play • Group discussions • Individual assignments • Demonstration • Field visit
ASSESSMENTMETHODS: • Written exam/test • Questioning • Demonstration and observation • Direct observation of tasks(real or simulated practice) using checklist • Review of tasks (portfolio, logbook, report, assignment…)

ASSESSMENT CRITERIA

Unit One: Health education and communication

- Describe concepts of health and health education
- Perform assessment and identify gaps
- Organize community and all available resources
- Identify target groups
- Prepare health education plan
- Design methods and approaches of health communication information dissemination
- Provide health education services
- Monitor service utilization and evaluate behavior change
- Develop, promote, implement and review for internal and external
- Maintain work related network and relationship
- Monitor service utilization and evaluate behavior change
- Describe approaches to meet communication needs

Unit Two: Train model families

- Identify better performing households
- Establish space and time for training
- Identify and collect resources for training
- Provide training according to MOH guidelines
- Provide post training follow up and monitoring
- Evaluate and certify well performing model households

Unit Three: Plan and Undertake advocacy on identified health issues

- Prepare advocacy plan to address health issues
- Consult community representatives' health needs and priorities
- Identify and consult influential community representatives and health development armies to disseminate IEC- BCC activities
- Plan, implement and evaluate of advocacy and community mobilization
- Organize and provide continuous advocacy services in partnership with stakeholders
- Use feedback for planning

ASSESSMENT CRITERIA:

Unit one: Basics of health care statistics

- Introduce health care statistics
- Describe variables and datasets
- Identify abbreviations, terminologies, and data specifications.
- Identify resources, limitations, and assumptions for dataset size.
- Categorize data

Unit two: Data collection in health care service

- Apply methods of data collection
- Use data collection tools
- Apply data collection process

Unit three: Data Organization and Storage

- Organize data
- Identify medium of data storage
- Structure and formulate storing and managing data
- Identify interval of categorical data
- Analyze variables

Unit four: Analysis of Health care Data

- Summarize data
- Analysis data
- Present data
- Identify degrees of uncertainty and assumptions
- Interpret data

Unit five: Computing HMIS indicators

- Introduce HMIS
- Describe HMIS indicators
- Define HMIS indicators
- Interpret HMIS indicators

Unit six: Key Performance indicators

- Apply KPIs for Hospitals and Health centers
- Formulate KPI

- Interpret KPIs

Unit Seven: Vital statistics

- Describe basics of Vital statistics
- Describe civil registration
- Compute and interpret vital statistics

Unit eight: HMIS information Utilization and Dissemination

- Target audiences for information dissemination.
- Apply methods for HMIS Presentation and dissemination
- Disseminate organizational Information
- Receipt output and feedback in HMIS
- Use HMIS information

Annex: Resource Requirements

HLT MLT3 M05 1224 Preparing Laboratory Solutions				
Item No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Learner)
A.	**Learning Materials**			
1.	TTLM		25	1:1
2.	Textbooks			
3.	Reference Books			
3.1	District laboratory practice in tropical countries. Part I	Monica Cheesbrough.second;2009	5	1:5
3.2	District laboratory practice in tropical countries. Part II	Monica Cheesbrough. Second; 2005	5	1:5
3.3	Medical Laboratory Technology A Procedure manual for Routine Diagnostic tests	Kani L Mukhjee, volume I, II,	5	1:5

		III 1988.		
3.4	Production of basic diagnostic Laboratory reagents	WHO Regional Office for the Eastern Mediterranean; 1995	5	1:5
3.5	Haematology for Medical Laboratory Technology Students; lecture note series.	Zewdnesh S. 2002.	5	1:5
3.6	Parasitology for Medical Laboratory Technology., Students lecture notes series	Girma M, Mohamed A, 1st edition; 2002.	5	1:5
3.7	Medical bacteriology for medical laboratory, Technology Students Lecture notes series	Abilo T. 1st edition; 2002.	5	1:5
4.	Journals/Publication/Magazines			
B.	**Learning Facilities & Infrastructure**			
1.	Lecture Room	5*5m	1	1:25
2.	Library			
3.	Laboratory room	5*10m	1	1:25
C.	**Consumable Materials**			
1.	Paper	A4	5rim	1:5
2.	Pencil	HB	5	1:5
3.	Pen	Ball point	5	1:5
4.	Common stock solution	Standard		
6.	Glove	standard		
7.	Filter paper	standard		
8.	Distilled water	standard		
9.	Gauze	standard		

10.	PPE material	standard		
D.	**Tools and Equipments**			
1.	Ph meters	standard		
2.	Balances	standard		
3	Water baths	standard		
4	Measuring cylinders, beakers, conical flasks, volumetric	standard		
5	Flasks and pipettes	standard		
6	Funnels	standard		
7	Fume cupboards	standard		
8.	LCD projector	standard		

LEARNING MODULE 6	Logo of TVET Provider
TVET PROGRAMME TITLE: **Medical Laboratory Techniques Level III**	
MODULE TITLE: Collecting and Processing Medical Samples	
MODULE CODE: HLT MLT3 M06 0222	
NOMINAL DURATION: 300 Hours	
MODULE DESCRIPTION: This module covers knowledge, skills and attitude required to apply concepts of physiology and anatomy of human, collect, handle, and transport and prepare samples for testing at work site or field using specified equipment and standard or routine procedures in a way that ensures the integrity of subsequent samples.	

LEARNING OUTCOMES

At the end of the module the learner will be able to:

LO1. Apply concept of physiology and anatomy

LO2. Prepare to collect samples

LO3. Collect and handle sample

LO4. Transport and handle sample

LO5. Receive and log sample

LO6. Distribute samples

LO7. Prepare sample for testing.

LO8. Maintain safe work environment

MODULE CONTENTS:

LO1. Apply concept of physiology and anatomy.

1.4 Identifying human physiology and anatomy

1.5 Identifying type and nature of samples

1.6 Identifying time of samples collection and collection sites

LO2. Prepare to collect samples

2.8 Identifying the purpose, priority and scope of the sampling request

2.9 Identifying hazardous site and reviewing safety procedures

2.10 Confirming type of samples, site of collection , time of collection and how to collect

2.11 Availing all necessary materials and monitoring stock.

2.12 Ensuring cleanliness of all items

2.13 Checking all items are available

2.14 Confirming sequences of handling and any permit requirements

2.15 Checking vehicle and communication devices

LO3.**Collect and handle sample**

3.1 identifying and organizing sample collection areas.

3.2 Removing covering and locks of secured devices.

3.3 Looking for advice if the required samples cannot be collected

3.4 Selecting and using the required sampling tools

3.5 Following sample collection procedures
3.6 Labeling information
3.7 Collecting the desired type and quantity of samples
3.8 Recording sample appearance, environmental conditions and any other factors
3.9 Maintaining sample integrity and confidentiality of information

LO4. Transport and handle sample

4.1. Confirming the number and nature of samples on arrival
4.2. Ensuring samples have been matched to request format
4.3. Applying requirements for sample transport
4.4. Identifying accompanying documents for any special needs
4.5. Completing required documentations
4.6. Packing samples in the specified transport containers
4.7. Maintaining sample integrity
4.8. Delivering samples to reception points
4.9. Maintaining confidentiality of information
4.10. Maintaining the states of transport containers

LO5. Receive and log sample

5.1 Confirming the number and nature of samples/items
5.2 Checking samples are matched with request forms
5.3 Completing required documentations.
5.4 Recording date and time of arrival
5.5 Entering all information into the Laboratory Information Management System (LIMS)/log sheet
5.6 Applying required document tracking mechanisms
5.7 Processing 'Urgent' test requests
5.8 Ensuring security and traceability of all information.

LO6. Distribute samples

6.1. Grouping samples requiring similar testing
6.2. Distributing samples to each laboratory department

6.3. Distributing request forms for data entry

LO7. Prepare sample for testing.

7.1. Performing physical separation of the samples
7.2. Performing chemical separation
7.3. Preparing sub-samples and back-up sub-samples
7.4. Labeling all sub-samples ensure traceability and stored in accordance with SOPs.
7.5. Distributing sub-samples to defined work stations to maintain sample integrity
7.6. Monitoring and controlling sample conditions before, during and after processing.

LO8. Maintain safe work environment

8.1. Establishing work practices and using PPE
8.2. Minimizing environmental impacts of sampling and generation of waste.
8.3. Cleaning equipment, containers, work area and vehicles
8.4. Avoiding hazards due to laboratory equipment.
8.5. Ensuring the safe collection of all hazardous wastes disposal
8.6. Cleaning splashes and spillages immediately
8.7. Segregating laboratory wastes with safety policy
8.8. Disposing wastes.
8.9. Reporting accidents and spillages

LEARNING METHODS:

- Lecture
- Demonstration
- Simulation
- Role-play
- Group discussion
- Exercise
- Individual assignment

ASSESSMENT METHODS:

- Written exam/test
- Practical assessment
- Questioning or interview

- Oral examination

ASSESSMENT CRITERIA:

LO1. Apply concept of physiology and anatomy.

- Concept of physiology and anatomy of human are identified,
- Type and nature of samples are identified
- Time of samples collection and collection sites are identified

LO2. Prepare to collect samples

- The purpose, priority and scope of the sampling request is Identified
- Site hazards are identified and enterprise safety procedures reviewed
- Type of sample, site of collection , time of collection and how to collect sample are confirmed
- All necessary materials are availed and its stock is monitored.
- Pre-use and cleanliness checks of all items are ensured
- All items are checked against given inventory and packed to ensure safe transport.
- Handling sequence and any permit requirements are confirmed
- Vehicle and communication devices are checked in working order
- Required transport containers and materials are checked in the vehicle.

LO3.Collect and handle sample

- Sample collection area are identified and organized.
- Security devices, such as locks and covers are removed as required.
- Advice is sought if the required samples cannot be collected or if procedures require modification.
- The required sampling tools equipment are selected and used in accordance with given procedures
- Sampling procedures are closely followed to obtain required samples and maintain their integrity.
- Labeling information is recorded in accordance with enterprise/legal traceability requirements.
- Desired type and quantity of sample are collected based on standard operating procedure

- Sample appearance, environmental conditions and any other factors that may impact on sample integrity are recorded, when required
- Sample integrity and confidentiality of information are maintained at all times#
- Samples/Items are delivered to each laboratory department in accordance with enterprise procedures

LO4. Transport and handle sample

- Confirm the number and nature of samples/items to be handled on arrival
- Ensure samples have been matched to request format
- Requirements are applied to the transport of samples and/or equipment
- Be alert laboratory personnel to any special needs are identified on documents accompanying the samples/ items
- Required documentation are completed at handling point
- Samples are packed in the specified transport containers and under the required conditions/on triple package /
- Sample integrity is maintained at all times
- Samples are delivered to reception point in accordance with enterprise procedures
- Confidentiality of information is maintained
- Vehicle is maintained according to enterprise requirements
- State of transport containers is maintained to ensure that are fit for purpose
- Enterprise procedures are followed for the cleaning/decontamination of equipment and vehicle as necessary
- Samples to the required collection point are delivered and all documentation completed to ensure traceability.

LO5. Receive and log sample

- Confirm the number and nature of samples/items to be received
- Samples are checked and matched with request forms before they are accepted.
- Required documentation are completed at handling point
- Date and time of arrival of samples at enterprise are record
- Samples are entered into the Laboratory Information Management System (LIMS)./log sheet

- Required document tracking mechanisms are applied.
- 'Urgent' test requests are processed according to enterprise requirements.
- Security and traceability of all information, laboratory data and records are ensured
- Pre-use and cleanliness checks of all items are conducted to ensure they are fit for purpose.

LO6. Distribute samples

- Samples requiring similar testing requirements are grouped
- Samples are distributed to each laboratory department maintaining sample integrity
- Request forms for data entry or filing in are distributed accordance with enterprise procedures.
- Check that samples and relevant request forms have been received by laboratory personnel

LO7. Prepare sample for testing.

- Physical separation of the samples is performed, as required
- Chemical separation of the samples is performed, as required
- Sub-samples and back-up sub-samples that are representative of the source are prepared
- All sub-samples are labeled to ensure traceability and stored in accordance with SOPs.
- Sub-samples are distributed to defined work stations maintaining sample integrity and traceability requirements.
- Sample conditions are monitored and controlled before, during and after processing.
- Defined preparation and safety procedures are followed to limit hazard or contamination to samples, self, work area and environment.

LO8. Maintain safe work environment

- Established work practices and PPE are used to ensure personal safety and that of others.
- Environmental impacts of sampling and generation of waste are minimized.
- All equipment, containers, work area and vehicles are cleaned according to enterprise procedures.

- Hazards due to laboratory equipment are avoided before storage.
- The safe collection of all hazardous for waste disposal is ensured
- Splashes and spillages are cleaned up immediately using appropriate techniques and precautions.
- All laboratory wastes are segregated in accordance with safety policy in accordance with waste disposal
- All wastes are disposed of in accordance with enterprise procedures
- Appropriate protective equipment is used to ensure personal safety when sampling, processing, transferring or disposing of samples.
- All accidents and spillages are reported to supervisor.

Annex: Resource Requirements

HLT MLT3 M06 1224 Collecting and Processing Medical Samples				
Item No.	**Category/Item**	**Description/ Specifications**	**Quantity**	**Recommended Ratio (Item: Learner)**
A.	**Learning Materials**			
1.	TTLM	• Prepared by the trainer	25	1:1
2.	Textbooks	-	25	1:1
3.	Reference Books		6	1:5
3.1	District Laboratory practice in tropical countries Part I	Monica cheesbrough.2nd; 2009	5	1:5
3.2	District Laboratory practice in tropical countries Part II	Monica cheesbrough.2nd; 2005	5	1:5
4.	Journals/Publication/Magazines			
B.	**Learning Facilities & Infrastructure**			
1.	Lecture Room	5*5m	1	1:25
2	Laboratory room	5*10 m	1	1:25
3.	Library	Standard (colleges	1	

		library)		
C.	**Consumable Materials**			
1.	Paper	A4	5rim	1:5
2.	Pen	Standard		
3	Pencil and rubber	Standard		
4	Graph paper	Standard		
5	Bucher paper	Standard	10	1:3
6	Marker	Standard	12 per pack	
7	Printer ink	Standard	4	
8	White board marker	Standard	15	
9	Plaster	Standard		
10	Gloves	Standard		
11	Swabs	Standard		
12	Urine cup	Standard	1	1:25
13	Stool cup	Standard	1	1:25
14	Coagulated test tubes	Standard	1	1:25
15	Anti-coagulated test tubes	Standard	1	1:25
16	Request paper	Standard		
17	Pasteur pipette	Standard		
18	Sputum cup	Standard		
19	Blood culture bottle	Standard		
20	70% alcohol	Standard		
21	Cotton	Standard		
22	Syringe with needle	Standard		
23	Vacutainer needle with holder	Standard		
24	Capillary tube	Standard		
25	Lancet	Standard		
26	Surgical blade	Standard		
27	Microscopic slide	Standard		
28	Scotch tape	Standard		

D.	Tools and Equipment			
1.	Computer	Lap top	1	1:25
2.	LCD projector	LCD Projector	1	1:25
3.	Printer	Laser Jet	1	1:25
4	Photocopy machine	Standard	1	1:25
5	Scanner	Standard	1	1:25
6	Back up	Standard	1	1:25
7	White board	110X80mm	1	1:25
8	Triple packaging	Standard	5	1:6
9	Ice box	Standard	2	1:15
10	Refrigerator	2-8^{o}C	2	1:15
11	Torniquet	Standard	5	1:5
12	Safety box	Standard		
13	Biohazard container	Standard		

LEARNING MODULE 7	Logo of TVET Provider

TVET-PROGRAMME TITLE: **Medical Laboratory Techniques Level III**

MODULE TITLE : Applying Computer and Mobile Health Technology

MODULE CODE : **HLT HES3 M07 1224**

NOMINAL DURATION : 100 Hours

MODULE DESCRIPTION : This module describes the knowledge, skills and attitude required to use new or upgraded technology. The rationale behind this unit emphasizes the importance of constantly reviewing work processes, skills, and techniques in order to ensure that the quality of the entire business process is maintained at the highest possible level through the appropriate application of new technology.

LEARNING OUTCOMES

At the end of the module the learner will be able to:

LO1. Start computer, system information and features

LO2. Navigate and manipulate desktop environment

LO3. Identify the existing Health technologies

LO4. Apply the functions of technology

LO5. Evaluate new or upgraded technology performance

MODULE CONTENTS:

LO1. Start computer, system information and features (theory = 16 hours + demonstration =16 hours)

1.1. Ergonomic requirements
1.2. Occupational Health and Safety (OHS)
1.3. Computer basic functions and features
1.4. Customize desktop configuration

LO2. Navigate and manipulate desktop environment (theory = 4 hours + demonstartion = 8 hours)

2.1 Access and use desktop icons.
2.2 Window functions and roles
2.3 Desktop shortcut.

LO3. Identify the existing Health technologies (theory = 16 hours + demonstration = 32 hours).

3.1. Computer operating systems
3.2. Internet browsers

3.3. Mobile technology

3.4. M Health

3.5. E Health

3.6. Basic computer skills

LO4. Apply the functions of technology (theory = 4 hours).

4.1. Mobile/Smart phones and tablets

4.2. Functions of technology

LO5. Evaluate new or upgraded technology performance (theory = 4 hours).

5.1 Features of new or upgraded equipment

5.2 Environmental considerations

5.3 Sources of information

5.4 Feedback

LEARNING METHODS:

- Lecture
- Demonstration
- Group discussion
- Exercise
- Individual assignment

ASSESSMENT METHODS:

- Interview
- Written test
- Demonstration/Observation

ASSESSMENT CRITERIA:

LO 1.Start computer, system information and features

- Workspace, furniture, and equipment are adjusted to suit user ergonomic requirements.
- Work organization is ensured to meet organizational and Occupational Health and Safety (OHS) requirements for computer operation.
- Computer is started or logged on according to user procedures.
- Basic functions and features are identified using system information.
- Desktop configuration is customized, if necessary, with assistance from appropriate persons.
- Help functions are used as required.

LO 2.Navigate and manipulate desktop environment

- Features are opened, closed, and accessed by selecting correct desktop icons.

- Desktop windows are opened, re-sized and closed by using correct window functions and roles.
- Shortcuts are created from the desktop, if necessary, with assistance from appropriate persons.

LO 3.Identify the existing Health technologies

- The existing knowledge and techniques to technology are applied
- Computer operating systems are utilized.
- Internet browsers are opened and manipulated to search for, send and receive information
- Situations are identified where existing knowledge can be used as the basis for developing new skills.
- Mobile technology skills are acquired and used to enhance learning and provision of standard health care
- Health techniques are used to enhance efficient utilization of resources and avoid duplication of efforts
- New and/or upgraded equipments are identified, classified and used where appropriate, for the benefit of customers as well as the health care system.

LO 4.Apply the functions of technology

- Mobile/Smart phones and tablets are used for solving organizational problems
- The functions of technology are applied to assist in solving the health and related data collection, organization, analysis and interpretation.
- Testing of new or upgraded equipment is conducted according to the specification manual.
- Features of new or upgraded equipment are applied within the organization
- Sources of information is accessed, used and interpreted relating to new or upgraded equipment

LO 5.Evaluate new or upgraded technology performance

- New or upgraded technology performance is evaluated and determined by introduced technology (mobile/ Mhealth, tablets)
- Mobiles/Smart phones and tablets are evaluated for the performance, usability and against the OHS standards
- Environmental considerations from new or upgraded equipment are determined.
- Feedback is used from appropriate performance evaluation.

Annex: Resource Requirements

HLT MLT3 M07 1224 Applying computer and Mobile Health Technology				
Item No.	**Category/Item**	**Description/ Specifications**	**Quantity**	**Recommended Ratio (Item: Learner)**
A.	**Learning Materials**			
1.	TTLM	• Prepared by the trainer	25	1:1
2.	Textbooks	-	25	1:1
3.	Reference Books	• National health policy • HMIS guidelines • Mobile manuals	10	1:3
4.	Journals/Publication/Magazines	• Health Indicators/latest • EDHS,2016 • Fact sheets • Standard formats	10	1:3
B.	**Learning Facilities & Infrastructure**			
1.	Lecture Room	5*5m	1	1:30
2.	Computer lab with internet access	5*5m	20 computer	1:2
3.	Library	Standard (colleges library)	1	
C.	**Consumable Materials**			
1.	Paper	A4	5rim	1:5

2.	Pen	Ball point	As required	
3	Pencil	HB		
4	Graph paper	Standard		
5	Bucher paper	Standard	10	1:3
6	Marker	Standard	12 per pack	
7	Printer ink	Laser Jet	4	
8	White board marker	Standard	15	
9	Plaster	Standard		
D.	**Tools and Equipment**			
1.	Computer	Lap top	1	1:25
2.	Software	DHIS 2, TenaCare,SDK tools		
3.	LCD projector	LCD Projector Sony	1	1:25
4.	Printer	HP Laser Jet	1	1:25
5.	Photocopy machine	Standard	1	1:25
6.	Scanner	Smart	1	1:25
7.	Back up	Smart	1	1:25
8.	White board	110X80mm	1	1:25
9.	Smart phones	Samsung	1	1:25

LEARNING MODULE 8	Logo of TVET Provider
TVET-PROGRAMME TITLE: **Medical Laboratory Techniques Level III**	
MODULE TITLE : Performing Parasitological Examination	
MODULE CODE : HLT MLT3 M08 0222	
NOMINAL DURATION : 250 Hours	
MODULE DESCRIPTION : This module covers knowledge, skills and attitude required to detect and differentiate the structure and stage of parasites using basic tests and procedures identified with the discipline of parasitological laboratory using microscope and other methods.	
LEARNING OUTCOMES At the end of the module the learner will be able to: **LO1**. Identify concept of human parasitology **LO2**. Process samples and associated request details **LO3**. Set up and uses Microscope **LO4.** Perform tests **LO5**. Maintain laboratory records **LO6**. Maintain a safe environment	
MODULE CONTENTS: **LO1. Identify concept of human parasitology** 1.6. Understanding concept of human parasitology 1.7. Identifying principle of host and parasite interaction 1.8. Differentiating life cycles and diagnostic stage of parasites 1.9. Identifying methodology of parasitological examinations	

1.10. Identifying Microscope set up and uses

LO2. Process samples and associated request details

2.7. Checking samples and request form

2.8. Sorting specimens

2.9. Rejecting samples and request forms with reasons for non-acceptance

2.10. Logging acceptable samples and requested forms

2.11. Processing samples as required by requested tests

2.12. Storing Samples and sample components appropriately

LO3. Set up and uses Microscope

3.4. Setting up the light path to optimize resolution

3.5. Selecting and filtering the appropriate objectives

3.6. Ensuring the lenses are made clean

3.7. Adjusting settings and alignment of the light path

LO4. Perform tests

4.8. Selecting authorized tests for the requested investigations

4.9. Performing quality control procedures

4.10. Conducting Basic parasitological tests

4.11. Recording results

4.12. Verifying results before releasing for clinician/client

4.13. Discussing with colleague when results are panic

4.14. Storing unused sample or sample components properly

LO5. Maintain a safe environment

5.5. Using PPE and established safe work practices

5.6. Cleaning up spills using appropriate techniques

5.7. Minimizing generation of wastes

5.8. Ensuring safe disposal of biohazardous materials and other laboratory wastes

LO6. Maintain laboratory record

6.4. Entering on report forms or into computer systems.
6.5. Updating instrument maintenance logs
6.6. Maintaining Security and confidentiality

LEARNING METHODS:

- Lecture
- Demonstration
- Group discussion
- Exercise
- Individual assignment

ASSESSMENT METHODS:

- Practical assessment
- Written exam/test
- Questioning or interview

ASSESSMENT CRITERIA:

LO1. Set up and use microscope

- Concept of human parasitology is understood
- Principle of host and parasite interaction are identified
- Life cycles and diagnostic stage of parasites are differentiated
- Methodology of parasitological examinations are identified
- Microscope set up and uses are identified

LO2. Process samples and associated request details

- Samples and request form details are checked before they are accepted
- Specimens are sorted according to tests requested, urgent status and volume
- Samples and request forms that do not comply with requirements to their source are returned with reasons for non-acceptance
- Acceptable samples are logged and forms requested applying required document tracking mechanisms
- Samples are processed as required by requested tests
- Samples and sample components are stored appropriately until ready for testing

LO3. Identify concept of human parasitology

- The light path is set up to optimize resolution
- The appropriate objectives are selected and filtered for the sample being examined
- Ensure that the lenses are made clean
- Settings and alignment of the light path are adjusted to optimize performance
- Sample is placed correctly on the stage

LO4. Perform tests

- Authorized tests that are indicated for the requested investigations are selected
- Quality control procedures are performed
- Basic parasitological tests are conducted according to documented methodologies,
- All results are recorded by noting any phenomena that may be relevant to the interpretation of results
- Results are verified before releasing for clinician/client
- Colleague is discussed when result interpretation is outside parameters of authorized approval
- Unused sample or sample components are stored for possible future reference, under conditions suitable to maintain viability
- Tested sample or sample components are stored according to organizational sample retention policy for retesting when requested.

LO5. Maintain a safe environment

- Established safe work practices and PPE are used to ensure personal safety and that of other laboratory personnel
- Spills are cleaned up using appropriate techniques to protect personnel, work area and environment from contamination
- The generation of wastes is minimized
- The safe disposal of biohazardous materials and other laboratory wastes are ensured in accordance with enterprise procedures.

LO6. Maintain laboratory records

- Entries are made on report forms or into computer systems, accurately calculating, recording or transcribing required data as required

- Instrument maintenance logs are updated, as required
- Security and confidentiality of all clinical information, laboratory data and records are maintained

Annex: Resource Requirements

HLT MLT3 M08 0222 Performing Parasitological Examination				
Item No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Learner)
A.	**Learning Materials**			
1.	TTLM	• Flip chart • Posters • Job aids	25	1:1
2.	Textbooks	• Training modules • Text books	25	1:1
2.1	Diagnostic Medical Parasitology.	Lynne Shore Garcia 2007, 5th edition		
3	Reference Books			

3.1	Essentials of Medical Parasitology	Apurba Sankar Sastry and Sandhya Bhat K.2014	10	1:3
3.2	Parasitology for medical and clinical laboratory professionals	John W. Ridley 2012		
3.3	Medical Parasitology for Medical Laboratory technology students	Awol M, Cheneke W. 2007		
3.4	Medical Parasitology: Lecture Notes for r Health Science Students.	Dawit Assafa, et al. 2004		
3.5	Clinical Parasitology: A Practical Approach.	Elizabeth A. Gockel-Blessing 2013, 2nd edition		
B.4.	Journals/Publication/Magazines			
	Learning Facilities & Infrastructure			
1.	Lecture Room	5*5m	1	1:25
2.	Library	Standard	1	
3.	Demonstration laboratory	5*10	1	1:25
C.	**Consumable Materials**			
1.	Paper	A4	5rim	1:5
2.	Pen		As required	
3	Pencil and rubber			
4	Graph paper			
5	Bucher paper	SinArtline	10	1:3
6	Art line marker		12 per pack	
7	Printer ink	HP Laser Jet	4	
8	White board marker	6 per pack	15	
9	Plaster	Rol3		
10	slide and cover slide	50 pk of 100 pcs each		
11	10% Formalin	standard	2 bottles	
12	Ether/Benzene	standard	2 bottles	
13	Zinc Sulphate/Acetate	standard	2 bottles	

14	Physiological Saline	standard	2 bottles	
15	Romanosky stains	standard	4 bottles	
17	AFB stain	standard	1 bottle	
18	Eosin stain	standard	1 bottle	
19	Lugols iodine	standard	1 bottle	
20	NaCl 10 g/l	standard	1 bottle	
21	Trichrom stain	standard	1 bottle	
22	Gauze	standard	2 roll	
23	Filter paper	standard		
24	Spatula and spoon	standard		
25	Stool cup	standard		
26	Aluminum foil	standard		
27	RDT kit	Standard		
28	Applicator stick	Standard	5 boxes	
29	Cotton swab	Standard	2 packs	
30	Oil immersion	Standard	2 bottles	
D.	**Tools and Equipments**			
1.	Computer	Lap top	1	1:25
2.	LCD projector	LCD Projector	1	1:25
3.	Printer	HP Laser Jet	1	1:25
4	Photocopy machine	Canon	1	1:25
5	Scanner	Smart	1	1:25
6	Back up	Smart	1	1:25
7	White board	110X80mm	1	1:25
8	Gloves	Standard		
9	Laboratory Benches	Standard		
10	Centrifuges	Standard	1	1:25
11	Microscopes	Standard	8	1:3
12	Staining Jars	Standard	5	1:5
13	Colour plate illustrations	Standard		
14	Basins	Standard	5	1:5

15	Dust bin	Standard	5	1:5
16	Container for sharp instruments	Standard	5	
17	Balance	Standard	2	1:13
18	Volumetric glass wares	Standard	10	1:3
19	Apron	Standard		
20	Overhead projector	Standard	1	1:25
21	White board	Standard	1	1:25
22	LCD projector	Standard	1	1:25
23	Parasitological slide films	Standard		
24	Slide projector	Standard	1	1:25
25	Audio-Visual materials	Standard		
26	Laboratory Coat	Standard	25	1:1
27	Goggle	Standard	25	1:1
28	Funnel	Standard	10	1:3
29	Wash bottle	Standard	10	1:3
30	Shaker	Standard	5	1:5

LEARNING MODULE 9	Logo of TVET Provider
TVET-PROGRAMME TITLE: **Medical Laboratory Techniques Level III**	
MODULE TITLE: Performing Urine and Body fluid analysis	
MODULE CODE: HLT MLT3 M09 0222	
NOMINAL DURATION: 240 Hours	
MODULE DESCRIPTION: This unit module covers knowledge, skills and attitude required to determine the type and quantity of different metabolic products in urine and body fluid, and identification of the different components of urine sediments using tests and procedures identified with the discipline of urinalysis laboratory.	

LEARNING OUTCOMES

At the end of the module the learner will be able to:

LO1. Identify concept of urinalysis and body fluid

LO2. Testing methodology of urinalysis and body fluid is identified

LO3. Perform the test

LO 4. Maintain laboratory records

LO5. Maintain a safe environment

MODULE CONTENTS:

LO1 Identify concept of urinalysis and body fluid

1.1. Identifying the concept of renal physiology and anatomy.
1.2. Recognizing formation and composition of body fluids
1.3. Identifying metabolic products in urine and body fluid.

LO2 Testing methodology of urinalysis and body fluid is identified.

2.1. Processing samples and associated request details .
2.2. Sorting specimens to the requested tests.
2.3. logging accepted samples and request forms
2.4. Applying required document tracking mechanisms
2.5. Recording accepted samples and request forms
2.6. Applying required document tracking mechanisms

LO 3 Perform the test

3.1 Assembling the required equipment, materials and systems.
3.2 Selecting authorized tests
3.3 Applying quality control procedures and conducting Individual tests.
3.4 Recording all results.
3.5 Discussing abnormal result to interpret
3.6 Verifying results
3.7 Retaining samples for retesting.

LO 4 Maintain laboratory records

4.1. Entering laboratory results and Information into available plate form

4.2. Maintaining instrument logs.

4.3. Maintaining records of urine received.

4.4. Maintaining security and confidentiality

4.5. Maintaining laboratory data and records.

LO 5 Maintain a safe environment

5.1 Utilizing the established work practices and PPE.

5.2 Decontaminating spills.

5.3 Minimizing the generation of wastes.

5.4 Ensuring the safe disposal of biohazard materials and laboratory wastes

LEARNING METHODS:

- Lecture
- Demonstration
- Simulation
- Group discussion
- Exercise
- •Individual assignment

ASSESSMENT METHODS:

- Written exam/test
- Practical assessment
- •Questioning or interview

LO1. Identify concept of urinalysis and body fluid

- Concept of renal physiology and anatomy are identified
- Recognize formation and composition of body fluids
- Metabolic products in urine and body fluid are identified
- Testing methodology of urinalysis and body fluid is identified.

LO 2. Process samples and associated request details

- Specimens are sorted according to tests requested, urgent status and volume
- Samples and request forms that do not comply with requirements are returned to their source with reasons for non-acceptance

- Accepted samples and request forms are logged, applying required document tracking mechanisms
- Samples are processed as required by requested tests
- Samples and sample components are stored appropriately until ready for testing

LO 3. Perform testing

- The required equipment ,materials and systems are assembled
- Authorized tests that are indicated for the requested Investigations are selected
- Individual tests are conducted according to documented methodologies (standards), applying required quality control procedures
- All results, including /noting any phenomena that may be relevant to the interpretation of results are recorded
- Colleague is discussed when result interpretation is outside parameters of authorized approval
- Results are verified before releasing for clinician/client
- Tested sample or sample components are stored according to organizational sample retention policy for retesting when requested

LO 4. Maintain laboratory records

- Entries on report forms or into computer systems/laboratory information system are made accurately, recording or transcribing required data as required.
- Instrument logs are maintained as required.
- Records of urine received are maintained.
- Security and confidentiality of all clinical information are maintained
- Laboratory data and records are maintained

LO 5. Maintain a safe environment

- Established work practices and PPE are used to ensure personal safety OHS and that of other laboratory personnel.
- Spills are cleaned up using appropriate techniques to protect personnel, work area

and environment from contamination. • The generation of wastes is minimized. • The safe disposal of biohazard materials and other laboratory wastes is ensured in accordance with enterprise procedures.

Annex: Resource Requirements

HLT MLT3 M09 1224 Performing Urine and Body Fluid analysis				
Item No.	**Category/Item**	**Description/ Specifications**	**Quantity**	**Recommended Ratio (Item: Learner)**
A.	**Learning Materials**			
1.	TTLM	• Prepared by the trainer	25	1:1
2.	Textbooks	-		
2.1	Graff's Textbook of Routine Urinalysis and Body Fluids	Lillian A. Mundt and Kristy Shanahan. , 2nd edition; 2011	5	1:5
3	Reference Books			
3.1	Fundamentals of Urine & Body Fluid Analysis	Nancy A. Brunzel 2018, 4th edition	5	1:5
3.2	Urinalysis and Body Fluids	Susan King Strasinger 2008, 5the edition	5	1:5
3.3	Urinalysis in Clinical Laboratory Practice	Alfred H. Free and Helen M. Free 2018, 2nd edition	5	1:5
4.	Journals/Publication/Magazines	• Fact sheets • Standard formats	10	1:3
B.	**Learning Facilities & Infrastructure**			
1.	Lecture Room	5*5m	1	1:25
2	Laboratory room	5*10 m	1	1:25
3.	Library	Standard (colleges library)	1	1:25
C.	**Consumable Materials**			
1.	Paper	A4	5rim	1:5
2.	Pen	Standard		
3	Pencil and rubber	Standard		
4	Graph paper	Standard		

5	Bucher paper	Standard	10	1:3
6	Marker	Standard	12 / pack	
7	Printer ink	Standard	4	
8	White board marker	Standard	15	
9	Plaster	Standard		
10	Gloves	Standard		
11	Test tube rack	Standard	10	
12	Distilled water	Standard	1	1:25
13	Lens Paper	Standard		
14	Urine strip	Standard	10 pack	
15	Test tube	Standard	50	
16	Microscopic Slide	Standard	50 pk of 100 pcs	
17	Cover slide	Standard	50 pk of 100 pcs	
18	WBC diluting fluid	Standard	2 bottles	
19	Urine Cup	Standard		
20	Wright stain	Standard	2 bottles	
21	Giemsa stain	Standard	2 bottles	
D.	**Tools and Equipment**			
1.	Computer	Lap top	1	1:25
2.	LCD projector	LCD Projector	1	1:25
3.	Printer	Laser Jet	1	1:25
4	Photocopy machine	Standard	1	1:25
5	Scanner	Standard	1	1:25
6	Back up	Standard	1	1:25
7	White board	110X80mm	1	1:25
8	Microscope	Binocular & Light	8	1:3
9	Centrifuge	Macro type	1	1:25
10	Heamocytometer	standard	15	1:2
18	Refractometer	standard	1	1:25
19	Urinometer	standard	2	1:15

20	Ph meter	standard	1	1:25

LEARNING MODULE 10	Logo of TVET Provider
TVET-PROGRAMME TITLE: **Medical Laboratory Techniques Level III**	
MODULE TITLE : **Applying Basic Health Statistics and survey**	
MODULE CODE : **HLT MLT3 M10 1224**	
NOMINAL DURATION : 80 Hours	
MODULE DESCRIPTION : This module describes the knowledge, skills and attitude required to apply basic health statistics and health survey methods to improve community health related activities	
LEARNING OUTCOMES **At the end of the module the learner will be able to:** **LO1** Prepare for the application of health survey **LO2** Undertake data collection **LO3** Compile, interpret and utilize health data **LO4** Prepare and submit reports **LO5** Take intervention measures accordingly	
MODULE CONTENTS: **LO1. Prepare for the application of health survey** 1.1 Definition of terms 1.2 Characteristics of health statistics 1.3 Scales of measurement 1.4 Basic principles of health statistics 1.5 Calculating rates and ratios 1.6 Basic principles of health survey **LO2. Undertake data collection** 2.1 Types of questionnaires 2.2 Preparing questionnaire 2.3 Pre-testing, modifying and amending questionnaire	

2.4 Training on data collection procedures
2.5 Equipment/materials for data collection
2.6 Informing members of community about data collection
2.7 Inviting community leaders on data collection process

LO3 Compile, interpret and utilize health data

3.1 Collecting health data
3.2 Analyzing health data
3.3 Maintaining health data base system.
3.4 Diagrammatic presentation of data
3.5 Maintaining steps of confidentiality according to prescribed procedures.
3.6 Collecting and updating vital events
3.7 Preparing and utilizing data

LO4 Prepare and submit reports

4.1 Preparing reports using standard reporting formats
4.2 Report dissemination
4.3 Communicating Updates and reportable diseases

LO5 Take intervention measures accordingly

5.1 Discussion with key stakeholders regarding the health problems
5.2 Identifying materials throughout the consultation process
5.3 Providing feedback
5.4 Making contributions to the health problem of the community
5.5 Collecting information and data for better intervention

LEARNING METHODS:

- Lecture
- Demonstration
- Exercise
- Individual assignment

ASSESSMENT METHODS:

- Interview
- Written test
- Demonstration/Observation

ASSESSMENT CRITERIA:

LO1 Prepare for the application of health survey

- Characteristics of health statistics are identified
- Scales of measurement are explained Basic principles of health statistics are applied
- Rates and ratios are calculated
- Basic principles of health survey are applied

LO2 Undertake data collection

- Types of questionnaires are identified
- Questionnaire is prepared and made available
- Questionnaire is pre-tested, modified and amended
- Necessary personnel are trained on data collection procedures
- The necessary equipment/materials are identified to execute data collection
- Members of community are informed about data collection dates and time
- Community leaders are invited to support data collection process

LO3 Compile, interpret and utilize health data

- Necessary health data are collected as per organizational guideline
- Information collected is classified or sorted out on the basis of a clear understanding of the purpose for maintaining the database system.
- Diagrammatic presentation of data is prepared
- Steps to maintain confidentiality is followed according to prescribed procedures are taken.
- Vital events are continuously and consistently collected and updated timely in accordance with organization procedures and guidelines
- Data are prepared and utilized according to prescribed procedures and guidelines

LO4 Prepare and submit reports

- Reports are prepared using standard reporting formats
- Reports are disseminated responsible bodies
- Updates and reportable diseases are communicated to concerned bodies according to prescribed procedures and guidelines.

LO5 Take intervention measures accordingly

- Discussions are made with key stakeholders regarding the health problems
- Briefing materials throughout the consultation process are provided to identify and clarify issues of interest/concern to stakeholders and own organization
- Feedback is provided to the team leader or work team on the results of the consultation process
- Positive contributions are made to activities that develop an understanding of the factors contributing to the health problem of the community
- Further information and data are collected when needed for better interventions

Annex: Resource Requirements

HLT MLT3 M10 1224 Applying Basic Health Statistics and survey				
Item No.	**Category/Item**	**Description/ Specifications**	**Quantity**	**Recommended Ratio (Item: Learner)**
A.	**Learning Materials**			
1.	TTLM	• Prepared by the trainer	25	1:1
2.	Textbooks	-	25	1:1
3	Reference Books			
3.1	Epidemiology: An introductory health statistics and research books	Kenneth J.Roman. Second edition; 2012	5	1:5
3.2	Modern Epidemiology	Kenneth J.Roman, Sander Greenland, and Timothy L.Lash. Second edition; 2008		
3.3	Biostatistics	Wayne W. Daniel.		

		Tenth edition; 2013		
3.4	Introduction to Biostatistics	Chap T. Le, Lynn E. Eberly. Second edition; 2016		
4.	Journals/Publication/Magazines	• Health Indicators/latest • EDHS,2016 • Fact sheets • Standard formats	10	1:3
B.	**Learning Facilities & Infrastructure**			
1.	Lecture Room	5*5m	1	1:30
2.	Library	Standard (colleges library)	1	
C.	**Consumable Materials**			
1.	Paper	A4	5rim	1:5
2.	Pen	Ball point	As required	
3	Pencil and rubber	HB		
4	Graph paper	Standard		
5	Bucher paper	Standard	10	1:3
6	Marker	Standard	12 per pack	
7	Printer ink	Laser Jet	4	
8	White board marker	Standard	15	
9	Plaster	Standard		
D.	**Tools and Equipment**			
1	Computer	Lap top	1	1:25
2	LCD projector	LCD Projector	1	1:25
3	Printer	Laser Jet	1	1:25
4	Photocopy machine	Standard	1	1:25
5	Scanner	Standard	1	1:25
7	Back up	Standard	1	1:25
8	White board	110X80mm	1	1:25

LEARNING MODULE 11	Logo of TVET Provider
TVET-PROGRAMME TITLE: **Medical Laboratory Techniques Level III**	
MODULE TITLE: Performing Community Mobilization and Provide Health Education	
MODULE CODE: HLT MLT3 M11 1224	
NOMINAL DURATION: **120 Hours**	
MODULE DESCRIPTION: This module aims to provide the trainees with the knowledge, skills and right attitudes attitude required to undertake health education, advocacy and community mobilization on identified health issues.	
LO1. Conduct health education and communication **LO2.** Train model families **LO3**. Plan and Undertake advocacy on identified health issues	

MODULE CONTENTS:

LO 1. Conduct health education and communication

1.12. Concept of health and health education

1.12.1. Introduction to Health education and Communication

1.12.2. Definition of terms

1.12.3. Concepts and principles of health education and communications

1.13. Performing assessment and identify gaps

1.13.1. Assessment techniques

1.14. Organizing community and all available resources

1.14.1. Community organizations

1.14.2. Available resources

1.15. Identifying target group

1.16. Preparing health education plan.

1.16.1. Planning health education program

1.16.2. Planning process

1.17. Designing methods and approaches of health communication

1.17.1. Health education methods

1.17.2. Health education approaches

1.17.3. Health education materials

1.17.4. Effective communication skills

1.18. Providing health education service

1.19. Monitoring of service utilization and evaluation of behavioral change

1.19.1. Health education models

1.19.2. Monitoring of service utilization

1.19.3. Evaluation of behavior change

1.20. Developing, promoting, implementing and reviewing strategies for internal and external dissemination of information

1.20.1. Disseminate relevant information

1.20.2. Relevant, policies and regulations

1.21. Maintaining work related network and relationship.

1.22. Approaches to meet communication needs

LO2. Train model families

2.7. Identifying better performing households

2.8. Establish space and time for training

2.9. Identifying and collecting resources for training

2.10. Providing training according to MOH guidelines

2.11. Provide post training follow up and monitoring

2.12. Evaluating and certifying well performing model households

LO3: Plan and Undertake advocacy on identified health issues

3.7. Preparing advocacy plan to address health issues

3.1.1. Stages of advocacy

3.1.2. Advocacy tools and approaches

3.1.3. Principles of effective advocacy

3.8. Consult community representatives health needs and priorities

3.9. Identifying and consulting influential community representatives and health development armies to disseminate IEC-BCC activities

3.10. Planning, implementation and evaluation of advocacy and community mobilization

3.11. Organizing and providing continuous advocacy services in partnership with stakeholders

3.12. Using feedback for planning

TEACHING METHODS:

- Lecture and discussion
- Role play
- Group discussions
- Individual assignments
- Demonstration
- Field visit

ASSESSMENT METHODS:

Competence may be accessed through:

- Written exam/test
- Questioning
- Demonstration and observation
- Direct observation of tasks(real or simulated practice) using checklist
- Review of tasks (portfolio, logbook, report, assignment…)

Assessment criteria

LO1: Conduct health education and communication

- Assessment and gap identification activities are performed according to organizational manual
- Community and all available resources are organized as per content requirement
- Target group identification is done according to organizational guideline
- Health education plan is prepared as per the requirements of target group organizational guideline.
- Methods and approaches of health communication are designed according to organizational manual
- Health education service is provided as per the requirements of target group
- Monitoring of service utilization and evaluation of behavioral change is noted in accordance with organizational manual
- Strategies for internal and external dissemination of information are developed, promoted, implemented and reviewed as required in accordance with workplace guideline
- Work related network and relationship are maintained as necessary.
- Different approaches are used to meet communication needs of clients and community.

LO2: Train model families

- Better performing household in their day to day activity is identified
- Space and time for training is established with consultation of appropriate personnel and community representatives
- Necessary resources are identified and collected as per the training plan
- Training is provided according to MOH guideline
- Follow up and monitoring are carried out in accordance with workplace guideline

- Well performing model house hold is evaluated and certified in accordance with workplace guideline

LO3: Plan and Undertake advocacy on identified health issues

- Advocacy plan is prepared to address an identified health issues as per organizational work guideline
- Community representatives are consulted to determine current health needs and priorities.
- Influential community representatives and health development armies are identified and consulted to disseminate IEC-BCC activities
- Continuous advocacy services are organized and provided in partnership with the stakeholders
- Feedback from community consultation and advocacy is used as a basis for planning

Annex: Resource requirement

Performing Community Mobilization and Provide Health Education HLT MLT 3 M11 1224				
Item No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Learner)
A.	**Learning Materials**			
1.	TTLM	• Prepared by the trainer	25	1:1
2.	Textbooks	-		
2.1.	The Pocket Guide to Health Promotion	Glenn Laverack, Open University Press, (2014)		1:5
2.2.	Promoting Health A Practical Guide	Angela Scriven, 2010 Elsevier Ltd. All rights reserved		1:5
2.3.	Lecture note Introduction to health education for health extension trainees in Ethiopia	Meseret Yazachew, Yihenew Alem, Jimma University (2004)		1:5
3.	Reference Books	National health policy Health education guidelines Health education modules		1:5
4.	Journals/Publication/Magazines	Health Indicators/latest EDHS,2016 Fact sheets Standard formats		1:5
B.	**Learning Facilities &**			

	Infrastructure			
1.	Lecture Room	5*5m	1	1:25
2.	Library	Standard (colleges library)	1	
3.	Demonstration room	Standard	1	1:6
C.	**Consumable Materials**			
1.	Paper	A4	5rim	1:5
2.	Pen	Standard		
3	Pencil and rubber	Standard		
4	Graph paper	Standard		
5	Bucher paper	Standard	10	1:3
6	Marker	Standard	12 per pack	
7	Printer ink	Standard	4	
8	White board marker	Standard	15	
9	Plaster	Standard		
D.	**Tools and Equipment**			
1.	Computer	Lap top	1	1:25
2.	LCD projector	LCD Projector	1	1:25
3.	Printer	Laser Jet	1	1:25
4	Photocopy machine	Standard	1	1:25
5	Scanner	Standard	1	1:25
6	Back up	Standard	1	1:25
7	White board	110X80mm	1	1:25

LEARNING MODULE-12	Logo of TVET Provider
TVET-PROGRAMME TITLE: **Medical Laboratory Techniques Level III**	
MODULE TITLE : **Applying 5S Procedures**	
MODULE CODE : **HLT MLT3 M12 1224**	
NOMINAL DURATION : 32 Hours	
MODULE DESCRIPTION : This module covers the knowledge, skills and attitude required to apply 5S techniques to his/her workplace. It covers responsibility for the day-to-day operations of the workplace and ensuring that continuous improvements of Kaizen elements are initiated and institutionalized.	
LEARNING OUTCOMES At the end of the module the learner will be able to: **LO1.** Develop understanding of quality system **LO2.** Sort needed items from unneeded **LO3.** Set workplace in order **LO4.** Shine work area **LO5.** Standardize activities **LO6.** Sustain 5S system	
MODULE CONTENTS: **LO1. Develop understanding of quality system** 1.1. Discussing quality assurance procedures 1.2. Understanding of quality system and continuous improvement 1.3. Identifying and relating workplace requirements 1.4. Explaining the 5S system **LO2. Sort items sort needed items from unneeded** 2.1 Identify items in the work area 2.2 Distinguish between essential and non-essential items 2.3 Sort items to achieve deliverables and value expected 2.4 Sort items required for regulatory or other required purposes 2.5 Placing any non-essential item appropriate place	

2.6 Checking regularly essential items are in the work area

LO3 Set workplace in order

3.1 Identifying the best location for each essential item

3.2 Assigning essential item in its location

3.3 Returning each essential item after use immediately

3.4 Checking regularly essential item in its assigned location

LO4. Shine work area

4.1. Keep the work area clean and tidy

4.2. Conduct regular housekeeping activities

4.3. Ensuring the work area is neat, clean and tidy.

LO5 Standardize activities

5.1 Follow procedures

5.2 Follow checklists for activities,

5.3 Keeping the work area to specified standard

LO6. Sustain 5S system

6.1 Cleaning up after completion of job

6.2 Identifying and Specifying situations

6.3 Specifying work area

6.4 Recommending improvements to lift the level of compliance

LEARNING METHODS:

- Lecture
- Demonstration
- Group discussion
- Exercise
- Individual assignment

ASSESSMENT METHODS:

- Practical assessment

- Written exam/test
- Questioning or interview

ASSESSMENT CRITERIA:

LO1. Develop understanding of quality system

- Discuss quality assurance procedures of the enterprise or organization
- Understand the relationship of quality system and continuous improvement in the workplace
- Identify and relate to workplace requirements the purpose and elements of quality assurance (QA) system
- Explain the 5S system as part of the quality assurance of the work organization

LO2. Sort needed items from unneeded

Identify all items in the work area

- Distinguish between essential and non-essential items
- Sort items to achieve deliverables and value expected by downstream and final customers
- Sort items required for regulatory or other required purposes
- Place any non-essential item in a appropriate place other than the workplace
- Regularly check that only essential items are in the work area

LO3 Set workplace in order

- Identify the best location for each essential item
- Place each essential item in its assigned location
- After use immediately return each essential item to its assigned location
- Regularly check that each essential item is in its assigned location

LO4. Shine work area

- Keep the work area clean and tidy at all times
- Conduct regular housekeeping activities during shift
- Ensure the work area is neat, clean and tidy at both beginning and end of shift

LO5.Standardize activities

- Follow procedures
- Follow checklists for activities, where available
- Keep the work area to specified standard

LO6. Sustain 5S system • Clean up after completion of job and before commencing next job or end of shift • Identify situations where compliance to standards is unlikely and take actions specified in procedures • Inspect work area regularly for compliance to specified standard • Recommend improvements to lift the level of compliance in the workplace

Annex: Resource Requirements

Item No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Learner)
A.	**Learning Materials**			
1.	TTLM	• Prepared by the trainer	25	1:1
2.	Textbooks	-	25	1:1
3.	Reference Books	• National health policy • IP guidelines • IP modules	10	1:3
4.	Journals/Publication/Magazines	• Health Indicators/latest • EDHS,2016 • Fact sheets • Standard formats	10	1:3
B.	**Learning Facilities & Infrastructure**			
1.	Lecture Room	5*5m	1	1:25
2.	Library	Standard (colleges library)	1	
3.	Demonstration room		1	1:6
C.	**Consumable Materials**			
1.	Paper	A4	5rim	1:5
2.	Pen	Standard		
3	Pencil and rubber	Standard		
4	Graph paper	Standard		
5	Bucher paper	Standard	10	1:3
6	Marker	Standard	12 per pack	

7	Printer ink	Standard	4	
8	White board marker	Standard	15	
9	Plaster	Standard	1	
10.	Labeling shelves	Standard	1	
11.	Cleaning agents (e.g., detergents, disinfectants)	Standard	1	
12.	marking areas	Standard	1	
13.	boxes	Standard	1	
14.	Bins	Standard	1	
15.	Colored tape	Standard	1	
16.	Posters	Standard	1	
17.	Gloves, goggles, masks, etc	Box	1	
D.	**Tools and Equipment**			
1.	Computer	Lap top	1	1:25
2.	LCD projector	LCD Projector	1	1:25
3.	Printer	Laser Jet	1	1:25
4	Photocopy machine	Standard	1	1:25
5	Scanner	Standard	1	1:25
6	Back up	Standard	1	1:25
7	White board	110X80mm	1	1:25
8.	Label Makers	Standard	1	
9.	Color-coded Tape	Standard	1	
10.	Signs and Posters:	Standard	1	
11.	Shelving Units	Standard	1	
12.	Vacuum Cleaners	Standard	1	
13.	Cleaning Cloths and Wipes	Standard	1	
14.	Calipers and Rulers	Standard	1	
15.	Floor Marking Paint	Standard	1	
16.	Personal Protective	Standard		

	Equipment (PPE)			
17.	Timers	Standard	1	
18.	Task Management Software	Standard	1	

Acknowledgement

The Ministry of Health and Ministry of labor and skill wishes to extend thanks and appreciation to the many representatives of industry, academe and government agencies who donated their time and expertise to the development of this Model Curriculum for the TVET program of Medical Laboratory Techniques Level III

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